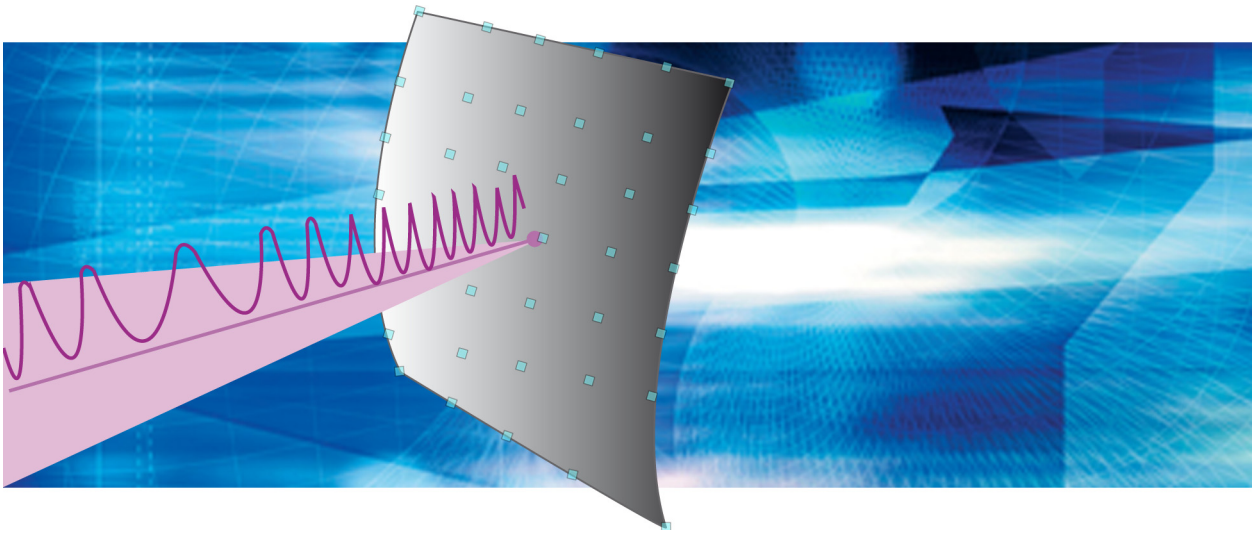
Polytec

Theory Manual

Polytec Scanning Vibrometer

PSV Theory



MSA

MSA-3D

PSV/PSV-3D

UHF

As of Software 9.1

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Appendix A: Introduction to Complex Numbers

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1 Introduction

Polytec's Scanning Vibrometers measure and analyze vibrations. If you are familiar with the theoretical background of signal processing and analysis, you can use the data acquisition system more efficiently. For further details refer to the specialist literature.

This manual explains the following:

- Harmonic vibrations - See CHAPTER 2 on this.
- Fourier transformation of the time signal into a frequency spectrum as well as further transformations of the calculated spectra - See CHAPTER 3 on this.
- Alias rejection - See CHAPTER 4 on this.
- Window functions - See CHAPTER 5 on this.
- Averaging - See CHAPTER 6 on this.
- Excitation signals - See CHAPTER 7 on this.
- Multiple reference analysis (MIMO) - See CHAPTER 8 on this.
- Digital filters to condition time signals - See CHAPTER 9 on this.
- VideoTriangulation - See CHAPTER 10 on this.
- Theoretical background of the bandwidth expansion of the UHF - See CHAPTER 11 on this.
- PC-based digital demodulation (VDD) - See CHAPTER 12 on this.
- How to optimize the measurement setup for the PSV - See CHAPTER 13 on this.
- How to work with 3D geometries - See CHAPTER 14 on this.
- Complex numbers - See APPENDIX A on this.

2 Description of Harmonic Vibrations

This chapter explains harmonic vibrations described with trigonometric functions as seen in SECTION 2.1 as well as harmonic vibrations described with complex numbers as seen in SECTION 2.2.

2.1 Harmonic Vibrations Described with Trigonometric Functions

Harmonic vibrations have a sinusoidal shape. A harmonic vibration is described by frequency, amplitude and zero phase angle as follows

$$U(t) = A \cdot \cos(\omega t + \varphi_0) \qquad \text{Equation 2.1}$$

with

$$\omega = 2 \cdot \pi \cdot f \qquad \text{Equation 2.2}$$

A… Amplitude
ω… Angular frequency
φ_0… Zero phase angle
f… Frequency

By means of the addition theorems, we can rewrite equation (2.1) as

$$U(t) = A\cos\varphi_0 \cdot \cos\omega t - A\sin\varphi_0 \cdot \sin\omega t \qquad \text{Equation 2.3}$$

or

$$U(t) = A_{cos} \cdot \cos\omega t - A_{sin} \cdot \sin\omega t \qquad \text{Equation 2.4}$$

$A_{cos} = A\cos\varphi_0$… Amplitude of the cosine part
$A_{sin} = A\sin\varphi_0$… Amplitude of the sine part

A harmonic vibration can thus be described as a superposition of a sinusoidal and cosinusoidal vibration whose amplitudes depend on the zero phase angle.

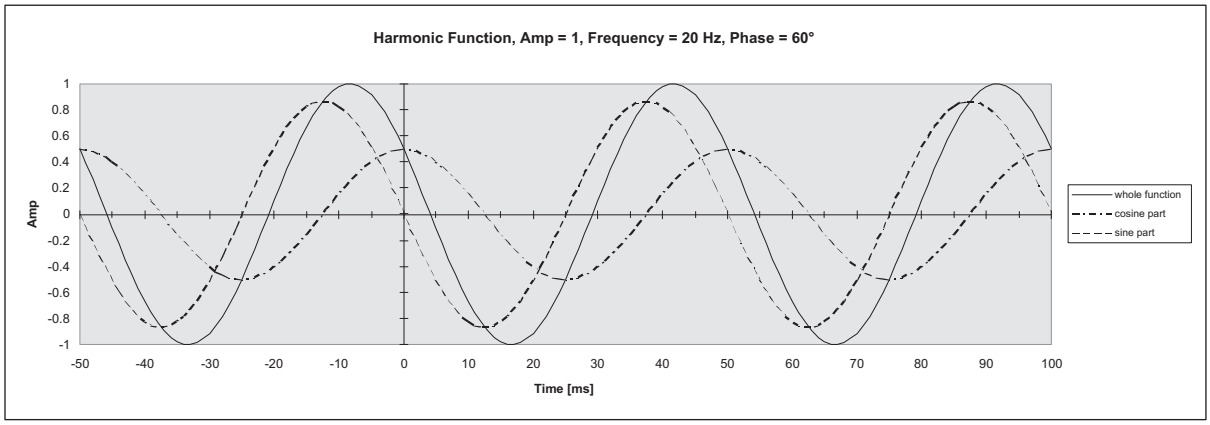


Figure 2.1: Harmonic vibration, split into cosinusoidal and sinusoidal part

2.2 Harmonic Vibrations Described with Complex Numbers

Instead of trigonometric functions, you can also use complex numbers to describe harmonic vibrations. This is advantageous for mathematical analysis and allows clear presentation in the complex number plane. If you are not familiar with complex numbers, you will find a short introduction to the calculus with complex numbers in APPENDIX A.

Complex numbers are underlined in this section.

With complex numbers, a harmonic vibration U(t) can be written as

$$\underline{U}(t) = Ae^{i(\omega t + \varphi_0)} \qquad \text{Equation 2.5}$$

The argument of the exponential function comprises a time-dependent part iωt and a constant part $i\varphi_0$.

The vibration can be made clearer in the complex pointer diagram. It corresponds to a complex pointer which revolves around the origin of coordinates with angular frequency ω.

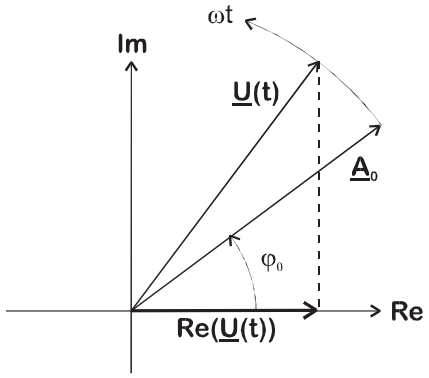


Figure 2.2: Vibration in the complex pointer diagram

The projection of the pointer to the real axis corresponds to the time-dependent instantaneous value of the real vibration.

$$\begin{aligned} Re[Ae^{i(\omega t + \varphi_0)}] &= ARe[\cos(\omega t + \varphi) + i\sin(\omega t + \varphi)] \\ &= A\cos(\omega t + \varphi) \end{aligned} \qquad \text{Equation 2.6}$$

Equation (2.5) can be rearranged as follows:

$$\begin{aligned} \underline{U}(t) &= Ae^{i\varphi_0}e^{i\omega t} \\ &= \underline{A}_0 e^{i\omega t} \end{aligned} \qquad \text{Equation 2.7}$$

$\underline{A}_0 = Ae^{i\varphi 0}$ … Complex amplitude

The complex amplitude contains information on the amplitude and the zero phase angle of the vibration.

3 Calculation of Transformed Signals

The software calculates various signals out of the recorded time signals. The different transformation steps are explained in the following sections. You can see a basic introduction into Fourier transformation in SECTION 3.1.

In addition, this chapter informs you on how the software calculates the following.

- Linear spectra - See SECTION 3.2 on this.
- Auto power and cross power spectra - See SECTION 3.3 on this.
- Frequency response and coherence - See SECTION 3.4 on this.
- Power Spectral Density (PSD) and Energy Spectral Density (ESD) - See SECTION 3.5 on this.
- RMS data - See SECTION 3.6 on this.
- 1/3 octave bands - See SECTION 3.7 on this.

3.1 Fourier Transformation

Periodic functions can be described as a sum of trigonometric functions (Fourier series). For example, a rectangular vibration can be described by the sum of the fundamental oscillation ($\sin\omega_0 t$ = 1st harmonic) and its odd harmonics. This results in

$$U_{rec}(t) = \sin\omega_0 t + \frac{1}{3}\sin 3\omega_0 t + \frac{1}{5}\sin 5\omega_0 t + \frac{1}{7}\sin 7\omega_0 t + \ldots \qquad \text{Equation 3.1}$$

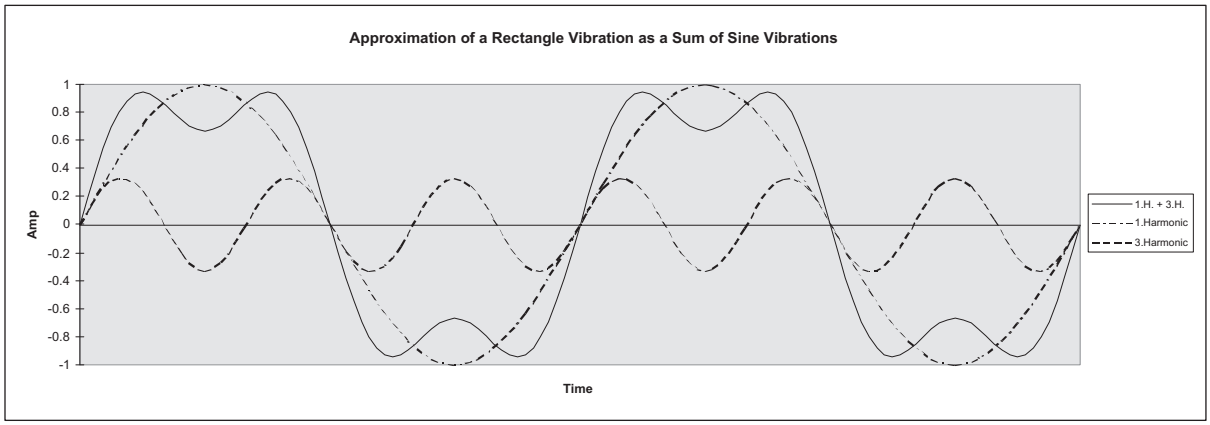


Figure 3.1: Fundamental oscillation (1st harmonic) superimposed with the 3rd harmonic

Non-periodic functions can be described as an integral of trigonometric functions (Fourier integral). An example is a single rectangular pulse.

The Fourier transformation splits any time signal into a sum of vibrations with different frequencies. All information which the time signal contains remains available. The original time signal is obtained by an inverse transformation.

3.1.1 Frequency Spectrum of a Rectangular Vibration

In the FFT spectrum of a rectangular vibration, lines appear at odd harmonics of the fundamental frequency f_0. This corresponds to splitting the rectangular vibration into sinusoidal vibrations. See equation (3.1) on this.

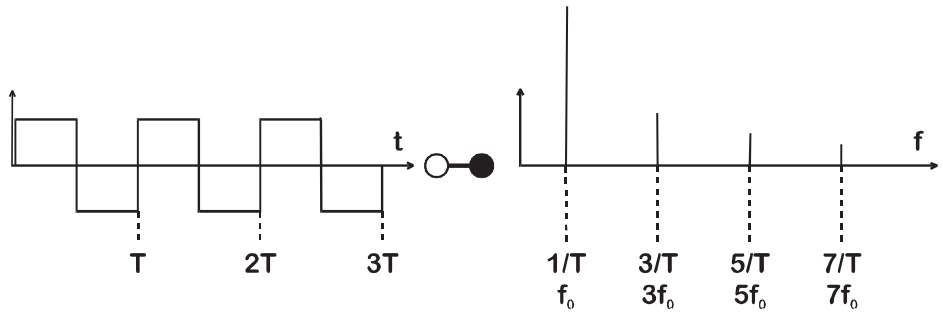


Figure 3.2: Rectangular vibration and its frequency spectrum

3.1.2 Frequency Spectrum of a Harmonic Vibration

A harmonic vibration with the frequency f_0 generates one line with frequency f in the FFT spectrum.

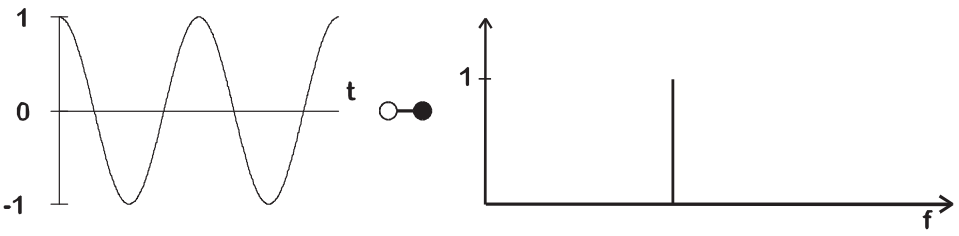


Figure 3.3: Harmonic vibration (Dirac Pulse) and its frequency spectrum

3.1.3 Frequency Spectrum of a Delta Function

The Delta Function is infinitely short and infinitely high. The integral of the Delta Function has the value 1. The frequency spectrum of the Delta Function has the value 1 at all frequencies.

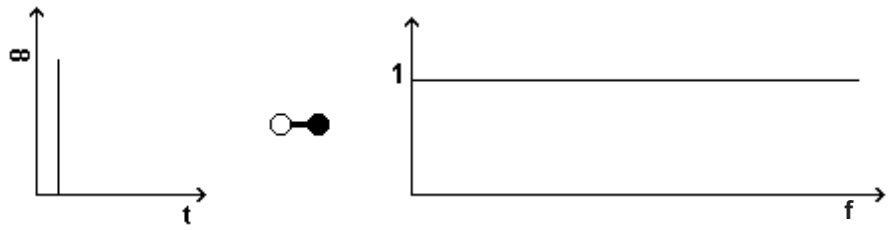


Figure 3.4: Delta Function and its frequency spectrum

The Delta function can not be physically realized, an approximation of the Delta Function is a hammer blow.

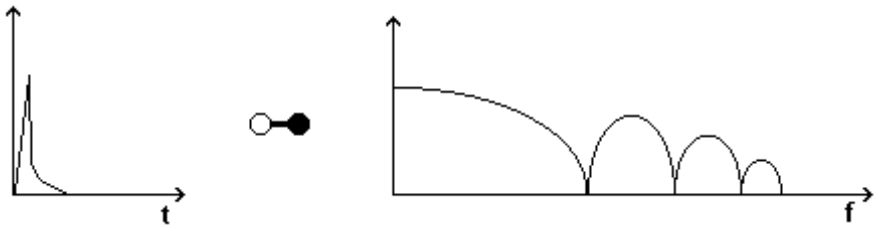


Figure 3.5: Hammer blow and its frequency spectrum

With a hammer blow an approximately constant excitation of each frequency takes place only at low frequencies. This must be considered when using a hammer blow to excite vibrations.

3.2 Generation of the Frequency Spectrum in the Software

The recorded time signal consists of a discrete number of samples. The software uses the FFT procedure (Fast Fourier Transformation) to generate the corresponding linear frequency spectrum.

Each frequency which appears in the time signal generates a spectral line in the FFT spectrum.

The number of samples which are processed by the FFT must be a power of two.

Scaling

For window functions Rectangle, Bartlett, Blackman-Harris, FlatTop, Hamming, Hanning and Tapered Hanning, the FFT is scaled in the software such that a harmonic vibration in the time signal with zero-to-peak amplitude 1 generates a line with amplitude 1 in the frequency spectrum if the harmonic vibration is periodic in the time window. Window functions Exponential and Force are not designed for harmonic signals. No additional scaling is done, the amplitudes of harmonic signals will be attenuated by these windows depending on the chosen window parameter. See CHAPTER 5 on this.

3.3 Auto Power and Cross Power Spectra

From the frequency spectra, auto power spectra (AP) and cross power spectra (CP) are calculated in the software by[1]

$$AP_{SS} = S^* \cdot S \qquad \text{Equation 3.2}$$

$$AP_{SS} = |S|^2 \qquad \text{Equation 3.3}$$

$$CP_{RV} = R^* \cdot V \qquad \text{Equation 3.4}$$

$$CP_{VR} = V^* \cdot R = CP_{RV}^* \qquad \text{Equation 3.5}$$

S … Spectrum at any channel

V … Spectrum of a vibrometer channel

R … Spectrum of a reference channel

* … Complex conjugate (refer to SECTION A.4)

3.4 Calculation of Frequency Response and Coherence

The frequency response of a vibrating system is the magnitude and phase response of the system at all frequencies of interest. The frequency response is defined as the Fourier Transform of the Impulse Response of a system.

Frequency response measurements require the excitation of the vibrating system with energy at all measured frequencies. The fastest way to perform the measurement is to use a broadband excitation signal that excites all frequencies simultaneously, and use FFT techniques to measure at all of these frequencies at the same time.

The coherence function is a measure of the power in the output signal caused by the input. If the coherence is 1, then all the output power is caused by the input. If the coherence is 0, then none of the output is caused by the input.

In the software the frequency response (FRF, H1, H2) and coherence functions (COH) are calculated in several steps. First FFTs are calculated of all active channels. See SECTION 3.2 on this. Then for each channel the auto power spectra (AP) and for every possible combination of a Vibrometer and reference channel the cross power spectra (CP) are calculated. See SECTION 3.3 on this.

Without averaging the FRF is calculated from the cross power and auto power spectra:

$$FRF_{VR} = \frac{CP_{RV}}{AP_{RR}} = \frac{V}{R} \qquad \text{Equation 3.6}$$

V … Spectrum of a Vibrometer channel (system output)

R … Spectrum of a reference channel (system input)

[1] All variables in this chapter (AP, CP, S, V, R, FRF, H1, H2, N) are functions of the frequency f. For better readability the argument (f) is omitted in the functions.

With averaging the response functions H1 and H2 and the coherence function COH are calculated:

$$H1_{VR} = \frac{\overline{CP}_{RV}}{\overline{AP}_{RR}} \qquad \text{Equation 3.7}$$

$$H2_{VR} = \frac{\overline{AP}_{VV}}{\overline{CP}_{VR}} \qquad \text{Equation 3.8}$$

$$COH_{VR} = \frac{H1_{VR}}{H2_{VR}} = \frac{\overline{CP}_{RV} \cdot \overline{CP}_{VR}}{\overline{AP}_{VV} \cdot \overline{AP}_{RR}} \qquad \text{Equation 3.9}$$

Here the overlined values denote averaged values. As the CP signals are always averaged in complex averaging mode and AP signals are real, H1, H2 and COH do not depend on the averaging mode.

H1 and H2 are estimators for the frequency response function FRF_{sys} of the measured system. While the phase is equal for H1 and H2, the magnitude differs. H1 is mainly affected by noise in the input signal (reference signal), whereas H2 is mainly affected by noise in the output signal (vibrometer signal).

ⓘ Without averaging H1 and H2 have the same result as FRF_{VR} in equation (3.6). The coherence has always the value 1. Therefore without averaging FRF is displayed instead of H1 and H2. The coherence is not displayed.

The following approximations can be made (the higher the number of averages, the better the approximation):

If $N^2_R \ll AP_{RR}$:

$$H1 = FRF_{sys}\left(1 - \frac{N^2_R}{\overline{AP}_{RR}}\right) \qquad \text{Equation 3.10}$$

If $N^2_V \ll AP_{VV}$:

$$H2 = FRF_{sys}\left(1 + \frac{N^2_V}{\overline{AP}_{VV}}\right) \qquad \text{Equation 3.11}$$

N^2_R… Power of the noise in the input signal

N^2_V… Power of the noise in the output signal

Here, N^2_R is the power of the noise in the input signal and N^2_V is the power of the noise in the output signal.

It can be seen that with the presence of input noise H1 is smaller than FRF_{sys} and with the presence of output noise H2 is bigger than FRF_{sys}.

As the coherence function is H1/H2, it can be approximated as:

$$COH = \frac{1 - \frac{N^2_R}{\overline{AP}_{RR}}}{1 + \frac{N^2_V}{\overline{AP}_{VV}}}$$

Equation 3.12

A further approximation for $N^2_R \ll \overline{AP}_{RR}$ and $N^2_V \ll \overline{AP}_{VV}$ results in:

$$COH = 1 - \left(\frac{N^2_R}{\overline{AP}_{RR}} + \frac{N^2_V}{\overline{AP}_{VV}} \right)$$

Equation 3.13

Example 1 A coherence value of 0.99 indicates that the sum of the relative input noise power and the relative output noise power $\left(\frac{N^2_R}{\overline{AP}_{RR}} + \frac{N^2_V}{\overline{AP}_{VV}} \right)$ is approximately 0.01. Coherence values close to 1 indicate a good signal-to-noise ratio.

Example 2 A coherence value of 0.1 indicates that the relative input noise power or the relative output noise power or both of them are high. The approximation formula is not valid anymore as $N^2 \ll AP$ is not fulfilled. Coherence values close to zero indicate a bad signal-to-noise ratio.

ⓘ If the number of averages is increased:

H1 converges to a smaller value than FRF_{sys}, depending on the input noise.

H2 converges to a bigger value than FRF_{sys}, depending on the output noise.

COH converges to a smaller value than 1, depending on the input and output noise.

For scanning measurements, normally the output (vibrometer) noise is bigger than the input (reference) noise, at least at some scan points. For visualizing an operating deflection shape (where all scan points are displayed simultaneously), H1 is the better choice than H2 because H1 is less sensitive to output noise.

3.5 Calculation of PSD and ESD

Power Spectral Density (PSD) is a measure used for random signals. Energy Spectral Density (ESD) is used for transient signals.

When measuring random signals, the amplitude level of the FFT depends on the set resolution bandwidth (RBW), which is dependent on the bandwidth, the number of FFT lines and the window function.

$$RBW = \frac{BW}{n_{FFT}} \cdot W_{RBW}$$ Equation 3.14

RBW… Resolution bandwidth

BW… Selected bandwidth

n_{FFT}… Number of FFT lines

W_{RBW}… Window factor for the resolution bandwidth (refer to TABLE 5.1)

The amplitude of the auto power spectra is proportional to the used resolution bandwidth. For a PSD spectrum, the auto power values are divided by twice the resolution bandwidth.

$$PSD = \frac{AP}{(2 \cdot RBW)}$$ Equation 3.15

When measuring transient signals, the amplitude levels of the FFT and the PSD depend on the set resolution bandwidth (RBW) and on the length of the time window. The ESD spectrum is scaled such that the amplitudes do not depend on the length of the time window. However, this requires that the transient signal completely dies out in the time window. For an ESD spectrum, the PSD values are multiplied by the length of the time window.

$$ESD = PSD \cdot t_W$$ Equation 3.16

t_W… Length of time window

3.6 RMS Calculation

RMS data is calculated out of the auto power spectrum.

$$RMS = \sqrt{\frac{\sum_{n=1}^{n_{FFT}} AP_n}{2 \cdot W_{RBW}}}$$

Equation 3.17

AP_n… Auto power spectrum

n_{FFT}… Number of FFT lines

W_{RBW}… Window factor for the resolution bandwidth (refer to TABLE 5.1)

This is equivalent to transforming the FFT data (of the linear spectrum) to the time domain and calculating the RMS value of the time data.

$$RMS = \sqrt{\frac{\sum_{n=1}^{n_{SAMPLES}} y^2_n}{n_{SAMPLES}}}$$

Equation 3.18

y_n… Measured values in time domain

$n_{SAMPLES}$… Number of samples

RMS values are calculated for the entire bandwidth and for all selected frequency bands. In the first case, any DC component is disregarded (n starts at 1), in the latter case only the FFT lines within the band limits are taken into account.

3.7 Calculation of 1/3 Octave Bands in the Software

1/3 octave bands are RMS values within pre-defined band limits. The display of 1/3 octave bands is also possible for signals calculated from a combination of channels (Vibrometer and Reference) like FRF, H1 and H2. Please note that in this case first the combined signals are calculated (refer to SECTION 3.4) and then the calculation of the 1/3 octave data is done.

Center frequency

For the center frequency f_m of the nth band we have

$$f_{m,n} = 10^{0,1n} \qquad \text{Equation 3.19}$$

n … Integer

For the output, the calculated values are rounded according to the German National Standard DIN EN 60651.

Band limits

With a logarithmic frequency scale, the bands limits are symmetric with respect to the center frequency. The following applies to the lower limit $f_{l,n}$

$$f_{l,n} = \frac{f_{m,n}}{10^{0,05}} \qquad \text{Equation 3.20}$$

and the upper limit $f_{u,n}$

$$f_{u,n} = f_{m,n} \cdot 10^{0,05} \qquad \text{Equation 3.21}$$

Display

On condition that a band contains at least two FFT lines, 1/3 octave bands are displayed approximately from the 8th FFT line onwards.

Integration and differentiation

Integration and differentiation of 1/3 octave bands are possible and are performed using the center frequency. Due to the fact that a 1/3 octave band contains multiple frequencies, there might be an error in the integrated or differentiated signal. The maximum theoretical error amounts to ±1.0 dB.

4 The Alias Effect

The software records and digitally processes analog time signals. Under certain conditions, the spectrum of digitally sampled signals may be falsified by the alias effect. In SECTION 4.1 alias rejection for the MSA and PSV is described. For background information on the alias effect, read SECTION 4.2.

4.1 Alias Rejection

All MSA and PSV systems provide built-in alias suppression. You do not need to take any measures to suppress aliases.

ⓘ For PSV-400: If you use the high-frequency decoder VD-05 or DD-300, alias rejection is only available for bandwidths above 4 MHz or sample frequencies above 10.24 MHz. With lower bandwidths or sample frequencies high-frequency signal portions and noise levels are not suppressed and interfere the measurement signal. In case of the DD-300 we recommend to use the output AUXILIARY UNIVERSAL on the vibrometer controller which has a bandwidth limited to 2 MHz and therefore the aliases will be suppressed.

4.2 Background Information on the Alias Effect

You can picture an actual analog time signal as a series of an infinite number of samples with an infinitely short distance. To digitally process analog time signals, they must be sampled for a certain period with a sampling frequency f_{Sample}. The number of samples becomes limited, which naturally means information is lost. It must therefore be made sure that the digital samples within the required accuracy provide the same information as the analog time signal.

If the time signal is sampled k times, the following applies:

$$Wt = \frac{k}{f_{Sample}}$$ Equation 4.1

Wt… Time window = length of the measured time signal

K… Number of samples

f_{Sample}… Sampling frequency

For correct digital signal processing, the sampling theorem must be fulfilled. This is guaranteed when the following applies:

$$f_{Sample} > 2 f_{Signal}$$ Equation 4.2

f_{Sample}… Sampling frequency

f_{Signal}… Maximum signal frequency

Signals with frequencies $f_{Signal} > f_{Sample}/2$ generate artificial lines at frequencies $f_{Alias} < f_{Sample}/2$. This is called the alias effect. The spectrum is falsified. To avoid infringement of the sampling theorem, signal frequencies which are higher than half the sampling frequency must be suppressed or the sampling frequency must be high enough. The suppression of frequencies above the chosen bandwidth is done automatically.

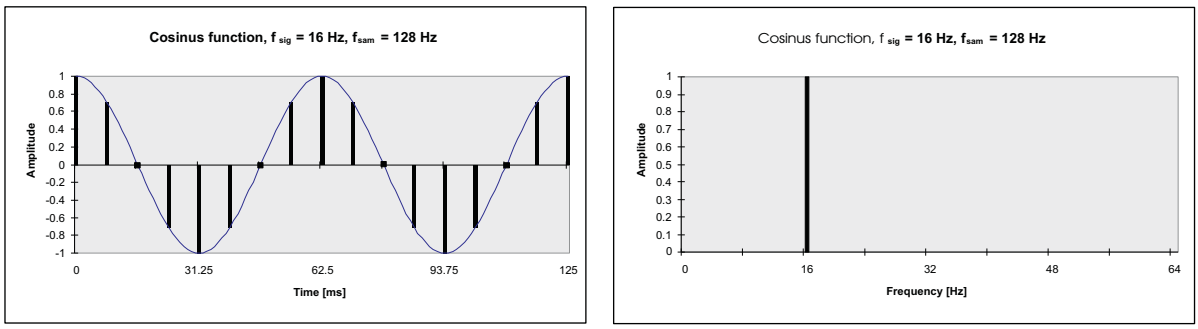


Figure 4.1: Sampling of a cosine function, sampling theorem fulfilled

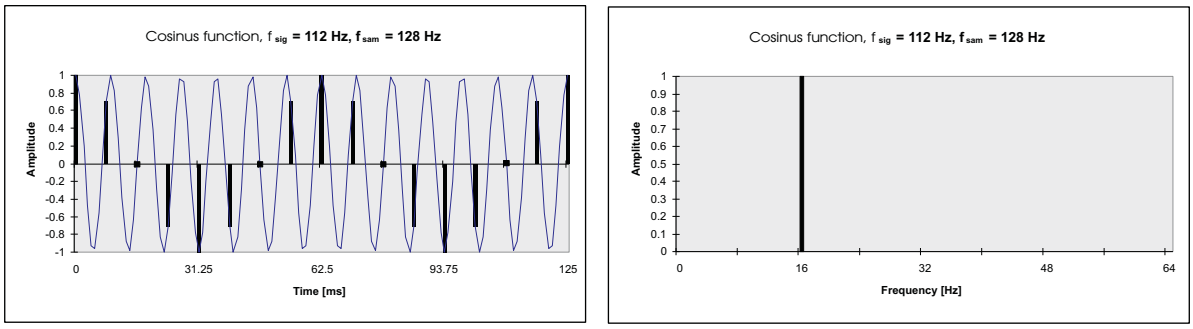


Figure 4.2: Sampling of a cosine function, sampling theorem infringed

Example Cosine function, sampled with K = 16 and f_{Sample} = 128 Hz.

With f_{Signal} = 16 Hz, a spectral line is generated at 16 Hz. With f_{Signal} = 112 Hz, the spectral line is also generated at 16 Hz. The spectra of f_{Signal} = 16 Hz and f_{Signal} = 112 Hz cannot be distinguished. The artificial spectral line (alias) with f_{Signal} = 112 Hz developed because the Nyquist theorem was infringed.

5 Window Functions

This chapter introduces you to the application of window functions in the software. In SECTION 5.1 you can find basic information on the effect of window functions in signal processing. SECTION 5.2 describes the parameters Attenuation and Ripple, while SECTION 5.3 shows the differences between Rectangular and Hanning Window. In SECTION 5.4 you will find an overview of the window functions in the software, and SECTION 5.5 contains the parameters of these window functions.

5.1 Basics

The result of the FFT is a discrete spectrum. Only frequencies which fall exactly on an FFT line are shown correctly in the frequency range. If there are frequencies in the time signal which do not fall on an FFT line, the information in the spectrum is distributed over the neighboring FFT lines. This is known as the leakage effect.

The leakage effect can be reduced by applying window functions. The time signal is weighted with the window function before calculating the FFT. For most window functions, the weighted time signal becomes zero at the start and at the end of the time window which thus avoids signal jumps at the edges of the time window.

The most simple window function is the rectangular window. The multiplication of a time signal with the rectangular window is synonymous with cutting off of the time signal outside the time window.

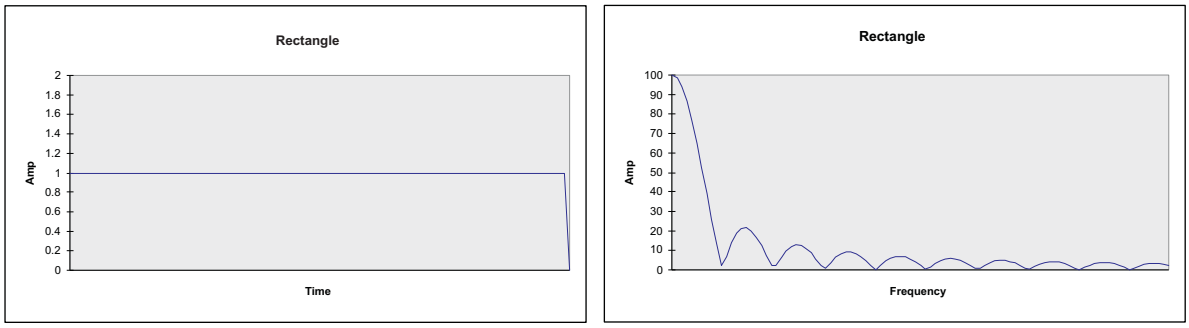


Figure 5.1: Rectangular window and its spectrum

The following applies to a periodic time signal:

- If the time signal is periodic in the time window, the periodic continuation of the windowed signal corresponds to the actual time signal. See FIGURE 5.3 on this (rectangle, signal without leakage).
- If the time signal is non-periodic in the time window, the periodic continuation of the windowed signal no longer complies with the actual time signal. The frequency of the signal does not correspond to an FFT line. Therefore, the signal is approximated by the closest FFT frequencies. See FIGURE 5.3 on this (rectangle, signal with leakage).

Generally, in actual signals the periodic continuation of the windowed signal is not the same as the actual time signal. To minimize leakage, most window functions artificially minimize the amplitudes at the beginning and the end of the time window. Thus the phase shifts are minimized in the periodic continuation of the windowed time signal. Hanning, Hamming, Blackman Harris, Bartlett and Flat Top are among these window functions.

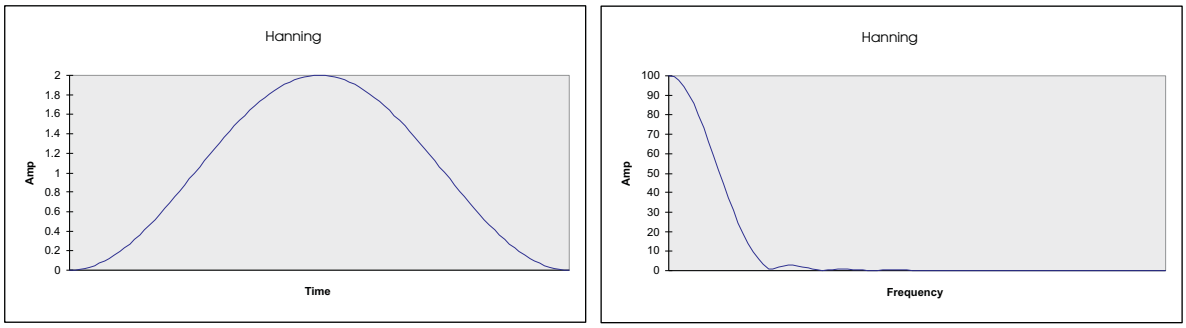


Figure 5.2: Hanning window and its spectrum

5.2 Attenuation and Ripple

All windows influence time signals which are non-periodic in the time window. A window function changes line shape and amplitude of the spectral lines in the frequency spectrum in actual measurements. Based on the spectrum of a window function, you can estimate how the window function influences the measured spectrum.

The spectrum of the rectangular window consists of a main maximum and secondary maxima. This applies to all window functions which artificially minimize amplitudes at the beginning and the end of the time window (Hanning, Hamming, Blackman Harris, Bartlett, Flat Top).

Attenuation

A measure of the frequency resolution of a window function is the frequency value by which a sharp spectral line in the windowed spectrum widens. A deciding factor is at which attenuation of the spectral line the line widening may appear.

If low attenuation is required, the frequency resolution depends on the width of the main maximum of the window spectrum. Half the width of the main maximum of the rectangular window (= $2\pi(f_{Sample}/K)$) serves as a means of comparison for the various window functions. The required attenuation is already achieved within the main maximum of the window spectrum and must be lower than the attenuation of the amplitude of the highest secondary maximum.

If higher attenuation is required, the frequency resolution depends on the amplitude of the highest secondary maximum in the window spectrum (highest side lobe attenuation). The required attenuation is achieved at the highest secondary maximum.

If very high attenuation is required, the frequency resolution depends on how fast the amplitudes of the secondary maxima decrease in the window spectrum (roll-off rate). The required attenuation is not yet achieved at the highest secondary maximum.

Normally, higher attenuation is required to clearly define spectral lines. The frequency resolution of a window function increases with the length of the time window.

With increasing length of the time window the following applies:

- The width of the main maximum decreases in the window spectrum.
- The amplitude of the secondary maxima in the window spectrum falls to lower values in most window functions as the window spectrum is longer.

The following applies:

$$Wt = \frac{K}{f_{sample}} > \frac{n_K}{\Delta f_0}$$ Equation 5.1

Wt…	Time window = length of the measured time signal
K…	Number of samples
f_{Sample}…	Sample frequency
n_K…	Index of the secondary maxima with required attenuation with regard to the main maximum
Δf_0…	Maximum permissible one-sided widening of a spectral line

All window functions are conceived for certain measurement tasks. With a window function which reaches a certain attenuation with a smaller n_K, you receive a higher frequency resolution with the same window width. Select a window function according to your measurement conditions.

Ripple

Ripple describes the quality of the reproduction of the amplitude of a window function. A windowed sinusoidal time signal with a frequency in the center of the bandwidth is observed. Ripple is the maximum attenuation of the frequency amplitude of this signal which is non-periodic in the time window. Windows with a very wide main maximum have the least Ripple. The frequency resolution decreases with increasing width of the main maximum.

5.3 Comparison of the Rectangular Window and the Hanning Window

The rectangular window does not falsify the spectrum of signals which are periodic in the time window. See FIGURE 5.3 on this.

The rectangular window strongly falsifies the spectrum of signals which are non-periodic in the time window.

The Hanning window falsifies the spectrum of signals which have the following properties:

- Periodic signal in the time window more than the rectangular window
- Non-periodic signal in the time window less than the rectangular window

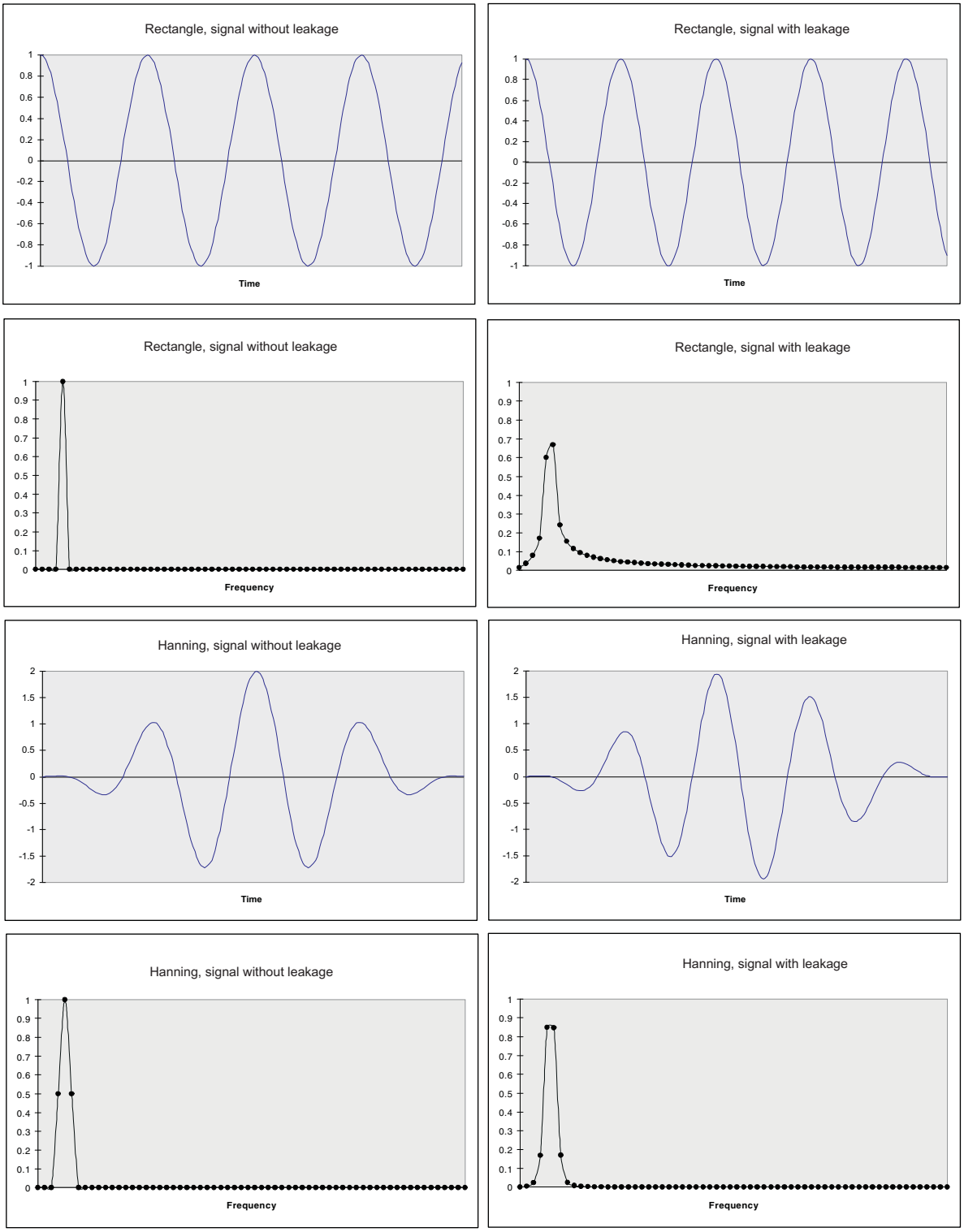


Figure 5.3: Comparison of the rectangular window with the Hanning window

5.4 Window Functions in the Software

The window functions which are implemented in the software are normalized as follows: A sinusoidal time signal which is periodic in the time window and with a frequency in the center of the bandwidth is considered. The rectangle window function reproduces the frequency amplitude of this signal in a pure manner. In order to also apply this to the window functions Hanning, Hamming, Blackman Harris, Bartlett and Flat Top, these window functions are multiplied with a scaling factor.

The individual window functions are described in detail in the following. In SECTION 5.5 you will find a brief overview of the most important parameters for the window functions implemented.

Rectangle

The main maximum in the spectrum of the rectangular window is the narrowest in comparison with other window spectrums.

Ripple: 3.92 dB, Highest side lobe attenuation: 13.3 dB.

The rectangular window is highly suitable for the following signals:

- Signals which are periodic in the window
- Non-periodic signals which decrease to zero within the time window (e.g. hammer blows)

The rectangular window is not suitable for higher and very high frequency resolutions, as the amplitude attenuation of the highest secondary maxima is low in the spectrum (highest side lobe attenuation is low) and the amplitudes of the secondary maxima only decrease slowly (roll-off rate is low).

Hanning

The Hanning window has the shape of an inverse, lifted cosine function.

With noisy measurement signals occurring in a noise excitation, a window function should display even-sized attenuating behavior in the complete window spectrum.

Ripple: 1.42 dB, Highest side lobe attenuation: 31.5 dB.

The Hanning window is highly suitable for strong noisy measurement signals as listed in the following:

- Amplitudes of the secondary maxima decrease rapidly (roll-off rate is high).
- Main maximum is only double the width compared to that in the rectangular window spectrum.
- Attenuation is constant.

Hamming

The Hamming window is a combination of the Hanning window and the rectangular window.

Ripple: 1.75 dB, Highest side lobe attenuation: 42.6 dB.

In comparison to the Hanning window, the Hamming window is more suitable for the frequency resolution of spectral lines in attenuations above the highest side lobe attenuation.

The Hamming window is not better suited than the Hanning window for very noisy measurement signals and the frequency resolution of spectral lines in an attenuation beneath the highest side lobe attenuation of the Hamming window, as the amplitudes of the secondary maxima only decrease slowly (roll-off rate is low) and the main maximum is double the width as compared to the rectangular window spectrum.

Blackman Harris

It is a good compromise for a window with hardly any influence on the line shape and amplitude of the spectral lines.

Ripple: 0.83 dB, Highest side lobe attenuation: 92.1 dB.

The Blackman Harris window is highly suitable for the following:

- The frequency resolution of strongly attenuated spectral lines, as the highest side lobe attenuation is very large and the secondary maxima decrease quickly at the beginning.
- Periodic measurement signals

As the main maximum in the Blackman Harris window spectrum is four times as wide as in the rectangular window spectrum, the Blackman Harris window is not better suited than for the following:

- The Hanning window for very noisy measurement signals as the attenuating behavior is very irregular in the Blackman Harris window spectrum.
- The frequency resolution of weakly attenuated spectral lines

Bartlett

The Bartlett window has the shape of a isosceles triangle.

The features of the Bartlett window lie between the rectangular window and the cosinusoidal windows Hanning, Hamming and Blackman Harris.

Ripple: 1.82 dB, Highest side lobe attenuation: 26.5 dB.

Flat Top

The Flat Top window is based on the cosine function. The main maximum in the Flat Top window spectrum is five times wider than in the rectangular window spectrum.

Ripple: 0.02 dB, Highest side lobe attenuation: 68.3 dB.

The Flat Top window is highly suitable for the following:

- A good reproduction of the amplitude height, e.g. for calibration purposes, as the ripple is very small
- Measurements with sinusoidal excitation

The Flat Top window is not suitable for the following:

- Very noisy measurement signals, as the attenuation is very irregular
- Measurements which require a detection of immediate spectral lines and a good reproduction of the line shapes of spectral lines

Exponential Unlike the other window functions, the exponential window does not attenuate signals at the beginning of the window.

The exponential window increases the attenuation proportionally. An exponentially decreasing signal is attenuated to the e-th part (e = 2.718…) of its initial value according to the time constant τ.

The exponential window is highly suitable for measurement signals which are excited with pulses (hammer blow) and are only attenuated weakly. With such measurement signals, the amplitude is high at the beginning of the time window and decreases slowly in an exponential manner. If the amplitude is not nearly zero at the end of the time window, the measurement is falsified. This effect is prevented by the exponential window and it receives the characteristic exponential decrease of the measurement signal.

The exponential window is not suitable for unattenuated periodic measurement signals.

5.5 Parameters of the Window Function

The most important parameters to distinguish between the window functions are the following:

- Maximum amplitude error (ripple)
- Highest side lobe attenuation
- Roll-off rate
- Increase of the resolution bandwidth.

You will find a summary of the values for these parameters in the following table.

Table 5.1: Parameters for window functions

Window function	Max. amplitude error %	Highest side lobe attenuation dB	Roll-off rate dB/dec	Window factor for the resolution bandwidth (W_{RBW})
Rectangle	–36	13	20	1.00
Hanning	–15	32	60	1.50
Hamming	–18	43	20	1.36
Blackman	–12	58	60	1.73
Blackman Harris	–9	92	20	2.00
Bartlett (triangle)	–19	27	40	1.33
Flat Top	–0.2	68	20	3.77
Tapered Hanning (Parameter = 10)	–35	12	20	1.08

The exponential and force windows have not been included in the table as it is not suitable for windowing continuous signals. It is used for impulse excitation if the answer signal does not decay in the time window. An important parameter for the exponential window is the time constant which can be set in the software.

6 Averaging

Measured signals are always superimposed by noise. To minimize the noise level, you can average in the software. This chapter describes averaging in time and frequency domains:

- Time domain averaging - See SECTION 6.1 on this.
- Frequency domain averaging - See SECTION 6.2 on this.
- Complex averaging - See SECTION 6.3 on this.
- Magnitude averaging - See SECTION 6.4 on this.
- Selecting a reference and trigger - See SECTION 6.5 on this.
- Peak Hold averaging - See SECTION 6.6 on this.

6.1 Time Domain Averaging

In time domain averaging a sequence of time traces is collected and averaged. A time trace is a contiguous sequence of samples. As time traces are acquired at different times and combined to the average result, the measurement conditions should be repeatable and the same for each time trace. This can be achieved for example by synchronizing the measurement to the excitation with the help of a trigger.

To use time domain averaging switch the acquisition mode to Time on the General page of the A/D Settings dialog, choose Time averaging mode and select the number of averages that contribute to the average result.

The time traces are collected one after the other and have the same number of samples. A single sample of the average result at time $t = t_0 + \Delta t$ after the first sample of the result is calculated from the samples at times $t_i = t_{0i} + \Delta t$ of the time traces i. Here t_{0i} is the start of time trace i. Therefore the average result has the same number of samples as the time traces. See the following figure for an illustration. The illustration shows that it is necessary to trigger the data taking of every time trace synchronized to the shown sine wave. Otherwise you would not get meaningful results from time domain averaging.

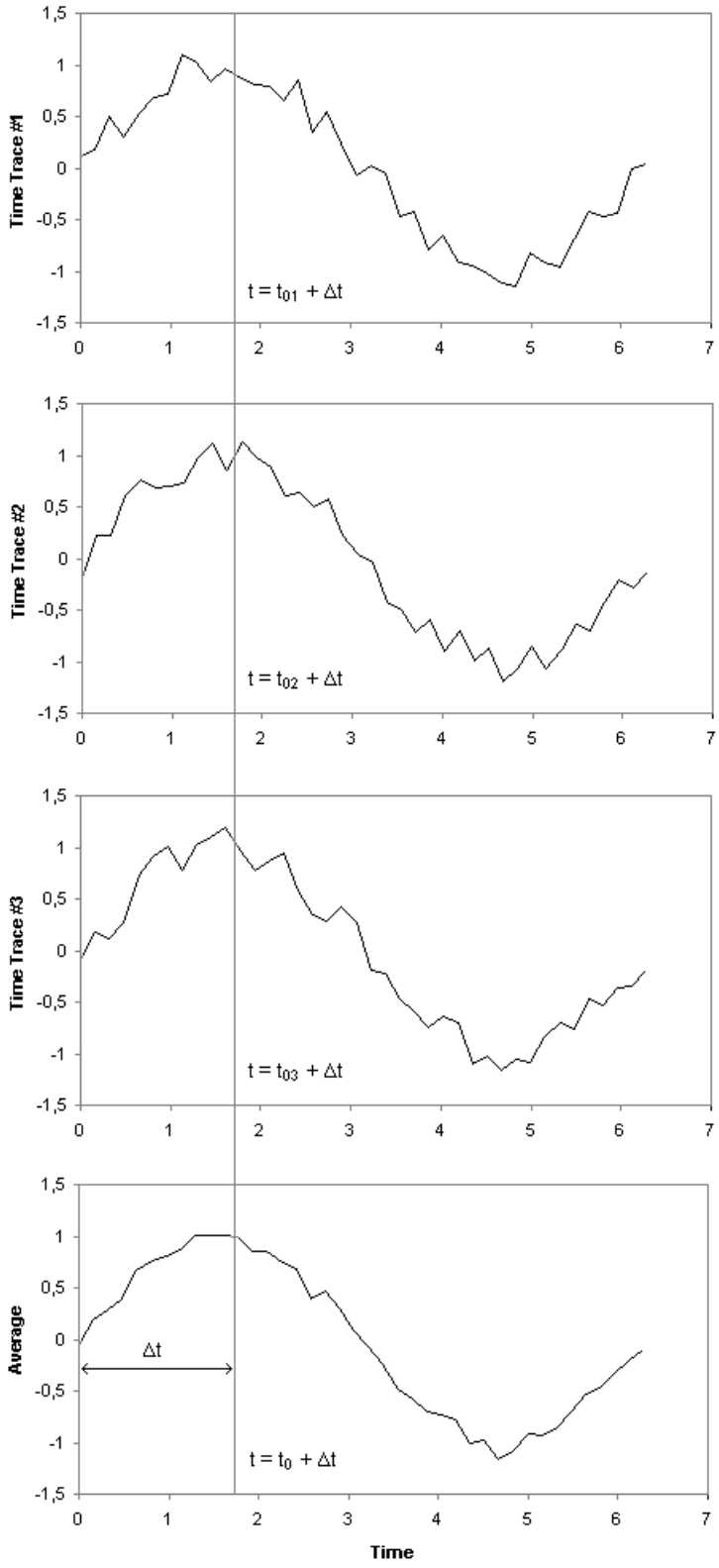


Figure 6.1: Principle of time averaging

There are two different modes available in time domain averaging: mean and median averaging. To select median averaging for an acquisition channel, tick the corresponding box on the SE (Signal Enhancement) page in the A/D Settings dialog.

In mean averaging mode the average is calculated by adding up the samples and dividing by the number of samples:

$$\bar{T} = \frac{1}{N} \cdot \sum_{n=1}^{N} T_n$$

Equation 6.1

In median averaging mode the samples at a given time after start of the traces are first sorted and then the middle value of the sorted values is used as the average value. If the number of traces is even, the mean of the two values adjacent to the middle is calculated. Median averaging mode is especially useful to suppress low probability, high amplitude noise, such as dropouts of the vibrometer signal.

The following table demonstrates the differences between median and mean averaging mode. Column A shows an example for 3 measurement values to be averaged. Both, the mean and median average give the same result. Column B shows the same sequence, except the middle value, which has been altered by a dropout in the vibrometer signal. The dropout clearly distorts the mean average, whereas it has little effect on the median average.

Trace/Sample	#1	#2
#1	3	3
#2	2	20
#3	4	4
Mean average	3	9
Median average	3	4

6.2 Frequency Domain Averaging

As in time domain averaging, in frequency domain averaging a sequence of time traces is collected. Each time trace has the same number of samples. From each time trace a spectrum is calculated by the means of an FFT. All spectra have the same number of FFT lines. The averaged spectrum is obtained by averaging all values at each frequency.

The following sections describe the different modes available in frequency domain averaging. There are some general rules for selecting complex or magnitude averaging mode for different applications:

If the phase between output and input signal is stable you can use complex averaging to reduce noise that is not phase correlated to the input signal. If the phase between output and input signal is not stable you can use magnitude averaging to obtain an average amplitude.

For example, complex averaging is used for deterministic excitations like a shaker driven with a periodic chirp signal whereas magnitude averaging is used for stochastic excitations like in a wind tunnel.

6.3 Complex Averaging

With complex averaging, N real parts and N imaginary parts of N measurement values are added and divided by N.

$$\bar{S} = \frac{1}{N}\sum_{n=1}^{N} S_n = \frac{1}{N}\cdot\left(\sum_{n=1}^{N} \mathrm{Re}(S_n) + i\cdot\sum_{n=1}^{N} \mathrm{Im}(S_n)\right)$$

Equation 6.2

This corresponds to the pointer in the following figure: $\frac{1}{2}\cdot(S_1 + S_2)$.

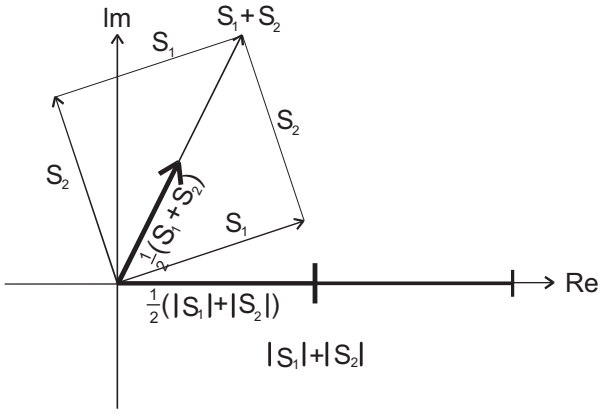


Figure 6.2: Complex and magnitude averaging of signals S1 and S2 (number of averages 2)

ⓘ If you perform measurements without a trigger the start of the measurement and therefore the phase is arbitrary. Nevertheless there is a fixed relation between the phase of the reference channel and the vibrometer channel. Prior to complex averaging the software relates the phase of all channels to that of the reference channel with the lowest number. See TABLE 6.1 on this.

6.4 Magnitude Averaging

With magnitude averaging, first the magnitudes of the complex values are calculated, then these N magnitude values are added and divided by N.

$$|\bar{S}| = \frac{1}{N} \cdot \sum_{1=n}^{N} |S_n|$$

Equation 6.3

This corresponds to the length of $\frac{1}{2} \cdot (|S_1| + |S_2|)$ in FIGURE 6.2.

ⓘ For cross power signals complex averaging is used even if magnitude averaging mode is selected on the General page in the A/D Settings dialog. See SECTION 3.4 on this. The method of averaging has no influence on the signals H1/H2/Coherence.

When using magnitude averaging, in the first instance all phase information is lost. To allow for animations of the data, the software assigns the phase of the cross power signal with the reference channel with the lowest number to the averaged result of the other channels. See TABLE 6.1 on this. This phase only makes sense if there is some phase correlation between reference and vibrometer channel.

6.5 Selecting a Reference and Trigger

If possible, you should always measure with a reference signal or trigger. Then you obtain useful phase information.

The phase relation of the vibrometer signal to the excitation is normally not constant at the individual scan points, but depends on the location of the scan points. Therefore the vibrometer signal must not be used as a trigger.

Reference As a reference signal, you can use the excitation of the vibration, for example.

Trigger As a trigger, you can use the following:

- The excitation itself
- A periodic or transient signal that comes from the object and that is in a fixed phase relation with the excitation

If the NO TRIGGER message appears in the status bar, the trigger signal is either missing or below the threshold (Level value, Acquisition Settings dialog, Trigger page).

1. In an analyzer, check the signal on which you are triggering.
2. Adapt the level to the signal with which you are triggering.

   The phase of the vibrometer signal is calculated by different methods, according to the measured channel, averaging and trigger as detailed in the following table.

Table 6.1: Calculation of the vibrometer signal

Averaging	Trigger	Ref available	Phase Vibrometer (V)	Phase Reference (R)
Off	Off	No	0	not available
Off	Off	Yes	Phase V referred to phase R	0
Off	On	No	Phase V referred to start of time window	not available
Off	On	Yes	Selectable[1]: Phase V referred to start of time window or Phase V referred to phase R	Selectable: Phase R referred to start of time window or 0
Magnitude	Off	No	0	not available
Magnitude	Off	Yes	Phase R*V (cross power spectrum) assigned after averaging	0
Magnitude	On	No	0	not available
Magnitude	On	Yes	Phase R*V (cross power spectrum) assigned after averaging	0
Complex	Off	No	settings not possible	settings not possible
Complex	Off	Yes	Phase V referred to phase R prior to averaging	0
Complex	On	No	Phase V referred to start of time window	not available
Complex	On	Yes	Selectable: Phase V referred to start of time window or Phase V referred to phase R prior to averaging	Selectable: Phase R referred to start of time window or 0

[1] Use the function Determine phase of the response channels from the frequency response function to the first reference channel on the Trigger page in the Acquisition Settings dialog.

You will find more detailed information in SECTION 6.3 and SECTION 6.4.

ⓘ If the phase referred to start of time window, the complete phase information is available for the inverse FFT (IFFT). Therefore, time signals can be reconstructed by the inverse FFT.

6.6 Peak Hold Averaging

Select Peak Hold for averaging if you want to calculate and display the respective maximum of the spectra over the set number of spectra.

The reference channel plays a special role with Peak Hold, as the measurement values of this channel decide which measurement values are to be used in the Peak Hold average spectrum for all channels. If several reference channels have been selected, then the one with the lowest channel number is used. If no reference channel has been selected, then the vibrometer channel with the lowest channel number is used. In the following, only the term reference channel is used.

During the measurement the spectra of the reference channel are analyzed. If, at a certain frequency, the magnitude of the current spectrum exceeds the maximum magnitude saved until now, then the measurement values of all channels at this frequency are saved as Peak Hold values. If there are complex measurement values, then they are also saved as complex values; however only the magnitude of the reference channel is used as the comparative criteria.

The way this algorithm works can be shown using as an example a Sweep excitation, which is connected to the reference channel. At the current Sweep frequency, the spectrum of the reference channel shows a peak. The magnitude of this peak is usually larger than that of all previous spectra. For this reason, the measurement values of all channels at this Peak frequency are copied into the Peak Hold average spectra. Once the Sweep has finished, the Peak Hold spectra from all channels are made up of data which corresponds to a peak magnitude of the reference.

This approach has several advantages. The Peak Hold spectra contain the required data, i.e. the data at which excitation was at a maximum. Apart from that, it is ensured that the measurement values from all channels at a given frequency come from spectra acquired at the same time. Therefore the H1, H2 and coherence signals can be calculated in the usual consistent way. See SECTION 3.4 on this. Furthermore, a brief drop out of the vibrometer signal in a single spectrum only affects the frequencies around the current Sweep frequency, but not the whole frequency range.

To further improve the signal quality, you can activate Signal Enhancement for the relevant channels. Then the combined signal quality of these channels is analyzed. The quality of the averaging spectra is compared to a threshold and as a result of this, the measurement values are either discarded for all channels or are used for calculating the Peak Hold spectra. In a sequence of measurements, the threshold automatically adapts to the average quality of the signals.

ⓘ If you select Peak Hold averaging, you cannot make any further quality adjustment for Signal Enhancement (Fast…Best).

If you want to use a Sweep excitation together with Peak Hold averaging, the sweep time should correspond to the measurement time. You can use the PeakHoldSettings.bas macro in the Examples folder to set all parameters appropriately.

Disable all reference channels if you want to perform the Peak Hold based on the signal of the vibrometer channel.

7 Excitation Signals

The quality of a vibration measurement depends strongly on the choice of the excitation signal. The principal criterion for the selection of the excitation signal is the purpose of the measurement. In case of data acquisition for modal analysis typically a complete spectrum must be measured at each scan point. Not all measurement tasks require a modal analysis. Questions can be answered much faster by measuring the deflection shapes at the dominant resonance frequencies, for example: Where on the structure is the highest vibration amplitude?

The linearity of the system also plays an important role for the selection of the excitation signal. If the device under test has non-linearities it is often desirable to get a linear approximation. This is particularly important for doing a modal analysis, as the used parameter estimation schemes are based on linear system models. Only certain excitation signals have the ability to average out non-linearities. With other signals non-linear distortions are clearly visible in the spectrum.

An important difference between excitation signals is the RMS-to-peak ratio, which is closely coupled to the obtained signal-to-noise ratio of the measurement. A higher RMS-to-peak ratio denotes a signal with higher energy content for a given peak amplitude. Generally, the higher the excitation energy, the lower the relative noise, at least insofar as the system behaves linearly.

Some excitation signals generate leakage effects in the spectrum calculated by FFT, which are an additional noise source. See CHAPTER 5 on this. However, a variety of signals exists which do not generate leakage effects intrinsically.

A further criterion is the reproducibility of the measurement. A scanning vibrometer measures all points sequentially. If the excitation is the same for all scan points and the test object and the test setup remain unchanged during the measurement, the sequential measurement is equivalent to a parallel one. If the excitation signal is exactly the same for each scan point, the vibrometer signal can be displayed as area data. If the excitation signal is different from scan point to scan point, the frequency response function must be used to visualize area data.

Available excitation signals

The signals available in the software can be grouped into four classes: harmonic, periodic in the time window, random and transient. Some signals belong to more than one class as shown in the following table.

Table 7.1: Available excitation signals and their classes

Signal	Class	RMS-to-peak	Leakage	Recommended window	Suitable for MIMO
Sine	Harmonic	0.7	no[1]	Rectangle[1]	no
Sweep	(Pseudo) Harmonic	0.7	yes	Hanning	no
Pseudo Random	Periodic	0.25	no	Rectangle	no
Periodic Chirp	Periodic	0.5	no	Rectangle	no
Burst Chirp	Transient	0.35	no[2]	Rectangle	no
White Noise	Random	0.25	yes	Hanning	yes
Burst Random	Transient, Random	0.2	no[2]	Rectangle	yes

[1] If frequency corresponds to an FFT line.

[2] If response dies out within the time window.

The following properties applies:

- Harmonic signals have the best signal-to-noise ratio and allow fast measurements at single frequencies.
- Signals which are periodic in the time window are best suited for fast measurements of complete spectra as they excite all frequencies simultaneously and do not generate leakage effects.
- If a linear approximation of a non-linear system is required, random signals must be used. Only with random signals non-linear distortions are averaged out.
- Transient signals excite a short vibration which dies out.The time window should be selected long enough to record the complete vibration signal until it died out. No leakage occurs.

ⓘ External generators do not offer all of the described signals nor does the internal generator of VibSoft-80 systems.

Sine

Sine excitation is used to measure the deflection shape at one particular frequency. The main advantages of Sine excitation are the very good signal-to-noise ratio and the possibility to perform a scan in a very short time. For scanning with sine excitation only the values at the excitation frequency need to be measured and saved. This can be done most efficiently using the least square method regression (FastScan), which is faster than an FFT and never suffers from leakage effects.

ⓘ Non-linearities can be detected by looking for harmonics in the spectrum. For that purpose measure in the FFT mode.

Sweep

Sweep denotes a sine signal, whose frequency is slowly altered, according to a ramp function. Slowly swept sine can be used for single point measurements, in combination with the Peak Hold function.

With Sweep excitation the resonance of a very lightly damped structure can be found which might remain undetected with other excitation signals.

The measurement time is proportional to the square of the frequency resolution. This means that normally the measurement is very slow and therefore Sweep is hardly ever used for scanning.

Periodic Chirp

Periodic Chirp is designed to excite all FFT lines of the measured spectrum. The time signal is generated out of the spectrum by an inverse Fourier transformation. Typically the magnitude is set for all frequencies to the same value. The phase is generated by an algorithm which maximizes the energy for a given maximum amplitude.

After waiting for steady state conditions the excitation and the response are measured without leakage effects. As all frequencies of interest are excited simultaneously no averaging is required. This is very useful in order to do fast measurements. However, for precise measurements averaging can be used in order to increase the signal-to-noise ratio.

The use of the rectangle window is required. Any other window will lead to major distortions.

Pseudo Random

Like Periodic Chirp, Pseudo Random is generated in the frequency domain. The only difference is the phase which is calculated as a uniformly distributed random number, giving the signal its random sound. The time signal is generated out of the spectrum by an inverse Fourier transformation. The properties of Pseudo Random are the same as for Periodic Chirp, the main difference is that the RMS-to-peak ratio is smaller for Pseudo Random.

ⓘ Although Pseudo Random has the word random in its name, it is no random signal. Once generated using random numbers for phase, it is repeated over and over again.

If a MIMO measurement has to be performed, Pseudo Random is not suitable as excitation signal. See CHAPTER 8 on MIMO measurements as well. Although a different random number sequence is generated for every channel, the principal component analysis cannot be performed, as those signals are repeated over and over again. Use Burst Random or White Noise excitation for MIMO measurements.

For linear systems the FRFs are the same for Periodic Chirp and Pseudo Random excitation. If measuring non-linear systems, Periodic Chirp and Pseudo Random both generate response signals with different distortions. Therefore, comparing the results with both excitation signals is a good method to check for non-linearity.

Burst Chirp

Burst Chirp is a fast swept sine signal followed by a zero signal. The data acquisition must be synchronized to the excitation by the internal trigger. In order to avoid leakage, the sample time (displayed on the frequency page) has to be chosen such that the response signal has decayed to zero at the end of the time window. Moreover the rectangle window has to be used.

To allow for time to reactivate the trigger, the complete chirp signal (including pause) is 20% longer than the selected time window.

For Burst Chirp the start and the end frequency is specified. To reverse the signal the start frequency can be set higher than the end frequency. The start (delay to SYNC) and duration of the chirp is specified in % of the sample time (displayed on the Frequency page). Burst Chirp is generated in the time domain.

Burst Random

Burst Random is a random signal followed by a zero signal. Each individual burst is a different random signal, which excites non-linearities differently for each single measurement. When using a sufficient number of averages (~25), a good linear approximation of the system can be obtained. Burst Random excitation is suitable for MIMO measurements. See CHAPTER 8 on this.

The data acquisition must be synchronized to the excitation by the internal trigger. In order to avoid leakage, the sample time (displayed on the Frequency page) has to be chosen such, that the response signal has decayed to zero at the end of the time window. Moreover the rectangle window has to be used.

To allow for time to reactivate the trigger, the complete burst signal (including pause) is 20% longer than the selected time window.

The start (delay to SYNC) and duration of the burst is specified in % of the sample time (displayed on the Frequency page). Burst Random is generated in the time domain.

White Noise

White Noise is a random signal with a flat spectrum adapted to the measured bandwidth. In difference to Pseudo Random, White Noise is not periodic in the time window but generated with a practically new sequence of random numbers over and over again.

White Noise is neither periodic nor does it allow the response to die out within the time window, therefore leakage occurs. In order to reduce the leakage effect, windowing – typically with the Hanning window – is applied.

To minimize leakage the frequency resolution has to be chosen sufficiently high, depending on the damping of the device under test.

As the input signal is always delayed by the measured system, only a part of the response correlates with the excitation. At the beginning of the sample period the response contains information caused by an excitation prior to the sample period. A part of the response excited in the sample period will occur only after the sample period. Both facts generate additional noise which can be reduced by a long sample time and averaging.

It is recommended to use the Hanning window and about 25 averages. The measurement can be sped up using 50% overlap, provided your system supports overlap. When using a sufficient number of averages (~25), a good linear approximation of the system can be obtained. White Noise excitation is suitable for MIMO measurements. See CHAPTER 8 on this.

Square, Triangle and Ramp

Those signals are available for special applications.

User defined

User defined signals are offered to allow for signals which are not implemented in the software.

8 Multiple Reference Analysis (MIMO)

In this chapter yo will find an explanation about Multiple Input Multiple Output (MIMO). See SECTION 8.1 on this. Its field of application is described in SECTION 8.2. You will find information on how to set up and analyze a MIMO test in SECTION 8.3. SECTION 8.4 describes the mathematics used by the PSV software to obtain the required results.

8.1 What is MIMO?

MIMO means Multiple Input Multiple Output and is a measuring technique where multiple inputs (excitations, references) are used to excite the measurement object at the same time and multiple outputs (responses) of the vibration are measured to obtain a matrix of FRFs between all excitation and response points that describes how the vibrating system reacts to an excitation.

8.2 Why Using Multiple References?

Setting up a multiple reference measurement is not an easy task and requires a good degree of careful preparation, understanding of the measurement and cross checks of the underlying assumptions to be successful.

Nevertheless multiple reference tests are an important tool for the following reasons:

- Using multiple inputs allows to put energy into the structure under test at several positions. Especially for large structures it is easier to provide enough energy with multiple inputs compared to a single input to excite global modes.
- If the structure under test exhibits closely coupled modes or repeated roots, the only way to insure that all modes can be identified is to measure with multiple references.

8.3 Example Application Using PSV

This section describes step by step the set up and analysis of a MIMO test using a scanning vibrometer. A typical application for MIMO is to find the modal shapes and mode properties of a large structure, such as a car body in white.

In a system with PSV-S-PCA option, multiple shakers can be used to excite the structure at different positions. With White Noise or Burst Random, an uncorrelated broadband random excitation generated by the output channels of the PSV's internal generator is used to drive the shakers. Generally we recommend to use a Burst Random signal combined with a Rectangle window function for MIMO tests. In the Acquisition Settings dialog on the Generator page, activate the Multiple Channels check box.

Force cells are used to measure the force fed into the structure. The force cells are connected to the reference channels of the PSV system. The channel properties are set according to the calibration factors of the cells. In the Acquisition Settings dialog on the Channels page, activate the check boxes of all reference channels with force cells in the Ref column. Deactivate the check boxes in the Ref column of all reference channels with response sensor (vibrometer, accelerometer).

At this time, you might want to check if the forces measured by the force cells are uncorrelated. To do so, proceed as follows:

1. In the Acquisition Settings dialog on the General page, select FFT mode, Magnitude averaging and the number of averages. See SECTION 6.4 on this. You have to provide at least M + 2 averages where M is the number of reference channels.
2. Activate the Principal Component Analysis (MIMO) box to let the software calculate the MIMO FRFs, principal inputs, multiple and virtual coherences during data acquisition.
3. Switch on the generator.
4. Start a continuous measurement in acquisition mode.
5. Have a look at the coherence signals between the various reference channels. The coherence signals should be near to zero over the whole frequency range, as the internal generator provides uncorrelated random signals on its three output channels. Nevertheless, the coupling of the shakers via the structure might lead to a remaining correlation between the driving forces. You might want to optimize the positions of the shakers to minimize this effect.
6. Have a look at the principal inputs signal. To do so, select the references channel. You will see three graphs (one for every reference channel). These graphs are the eigenvalues of the auto power/cross power matrix of the reference channels for every single frequency line. For every frequency the eigenvalues are sorted by size, so that graph one always contains the largest eigenvalue and the last graph contains the smallest eigenvalue. Therefore there is no direct link to the reference channel. Nevertheless you can see, if one or more graphs will show values that are near to zero and are very small compared to the others. Please note that this might be the case for a range of frequencies only. This indicates, that some of the references are correlated. Again you might want to optimize the positions of the shakers.

Although PSV offers just a single response channel, scanning the structure with the laser provides fast and efficient means to perform a sequence of MISO (multiple input, single output) measurements with a roving output. Due to the steady state condition and because of the inputs are measured at every scan point, these MISO measurements can be combined to one MIMO data set.

7. You can set up the geometry of the scan points to cover the parts of interest of the structure to the desired resolution. If there are unused reference channels, you can measure the responses in the excitation points (DOFs) by accelerometers to obtain the driving point function. Those accelerometer signals are response signals, therefore the Ref check box must not be activated for the corresponding channels. You can enter the indices of the driving points as reference point indices of the reference channels on the Channels page in the Acquisition Settings dialog. You should also set the direction of the forces and the measured vibration for every active channel on this page according to the geometry of your test set up. If you do not work with 3D geometries we recommend to enter the distance from the scanning head to the object at this time if you plan to export the data for further modal analysis (see below).
8. Start the scan.
9. After the scan has finished you can use the PSV presentation mode to analyze the deflection shapes at the resonances of the structure. You might want to export the FRFs and analyze them further with a modal analysis software package like Vibrant's ME'Scope to find the modal parameters and shapes of the structure.

   As you have set the reference number indices, the directions of the inputs and outputs and eventually the distance to the object before starting the measurement, no additional information has to be entered at the time of export apart from the choice which data to export.

8.4 The MIMO Model

This section describes the dynamic model that is used by PSV to obtain the FRFs, principle inputs and coherence signals during a MIMO test. The MIMO model is a linear frequency domain model where spectra (Fourier transforms) of multiple inputs are multiplied by elements of an FRF matrix to yield spectra of multiple outputs. The FRF matrix model can be written as:

$$V_N(f) = H_{NM}(f) \cdot R_M(f) \qquad \text{Equation 8.1}$$

M… Number of inputs (references)

N… Number of outputs (responses)

f… Frequency

$V_N(f)$… Vector of outputs

RM(f)… Vector of inputs

$H_{NM}(f)$… Matrix of the FRFs describing the system

If $H_{NM}(f)$ is known, the response of the system to a set of inputs can be predicted.

$H_{NM}(f)$ is a result of the type of MIMO test described above, where N is the number of scan points and M the number of the reference channels.

To obtain $H_{NM}(f)$, the software performs the following steps at every scan point:

1. A series of time traces is acquired. The number of traces corresponds to the number of averages specified. The signals at the response channel (vibrometer channel) and the reference channels are acquired simultaneously.
2. The window functions are applied to the time traces and a FFT (fast Fourier transform) is calculated for every channel and for every trace.
3. The auto power signals of all channels and the cross power signals of all combinations of response and reference channels are calculated and averaged over the spectra. For the cross power signals complex averaging is used (see SECTION 3.4).
4. The eigenvalue of the M by M matrix of the auto powers and cross powers $R_M(f) \cdot R_M(f)^*$ are calculated for very frequency f and sorted by size. These eigenvalues are stored as the principal inputs signal in the references channel.
5. The principal component analysis is performed on this set of cross power and auto power signals to obtain the FRFs of this scan point:

$$H_{iM}(f) = (V_i(f) \cdot R_M(f)^*) \cdot (R_M(f) \cdot R_M(f)^*)^{-1} \qquad \text{Equation 8.2}$$

i… Index of the current scan point

$V_i(f)$… Response measured by the vibrometer at this point

$R_M(f)$… Vector of the references measured

R*… Transposed complex conjugate of the vector R

$V_i(f) \cdot R_M(f)^*$ is the 1 by M vector of the cross powers between the responses and references and $(R_M(f) \cdot R_M(f)^*)^{-1}$ is the inverted M by M matrix of the auto powers and cross powers of the references.

6. The coherence signal is calculated for the current scan point. The coherence is a measure of the input energy visible at the output of the system. The software calculates two types of coherences, the virtual coherences and the multiple coherence. The multiple coherence is available for every response channel and is common for all reference channels. Only this signal can potentially reach a value of 1. The virtual coherences between the output channel and the principal inputs do in general never reach a value of 1. The Multiple Coherence is the sum of the virtual coherences over all reference channels. You will find both signals in the corresponding response channel. The virtual coherences signal contains M graphs. The first graph is the coherence between the response channel and the first principal input, i.e. the largest eigenvalue of the auto power/cross power matrix of the reference channels. The last graph is the coherence between the response channel and the smallest eigenvalue, respectively. The virtual coherences can be used to check how the energy put into the system is spread over the object. There might be parts of the object or frequency ranges, where the effective number of references is smaller than M. In this case some of the virtual coherences are near to zero and very small compared to others. In this case you might want to optimize the positions of the points where you put the energy on to the system.
7. The coherences between the input channels are calculated. This allows for checking the correlation between the inputs. These coherences should be near to zero.

After the scan has finished, all elements of the FRF matrix H_{NM} (f) are available for analysis.

To be able to solve equation (8.2), the following conditions have to be met:

1. The R_M inputs have to be uncorrelated between channels.
2. The R_M inputs have to be uncorrelated between time traces that contribute to the averages.
3. For all R_M inputs and all $V_N(f)$ outputs the same window function has to be used.
4. The number of averages has to be at least M + 2.

When you activate the Principal Component Analysis (MIMO) check box on the General page in the Acquisition Settings dialog and you are using the internal generator, the software automatically checks the above conditions.

ⓘ The software is not limited to a single output channel. You could use two of the acquisition channels as output and two as input by ticking the corresponding check boxes in the Ref column on the Channels page in the Acquisition Settings dialog. The software then correctly calculates the matrix $H_{22}(f)$ and the two multiple coherences of the output channels. This is the case with the PSV-3D where you measure three outputs simultaneously.

9 Digital Filters

Digital filters are used to condition time signals, predominantly if the evaluation is done in the time domain. As spectra are calculated from the time signals, digital filtering always also changes the spectra. If spectra are all that is to be evaluated, then as a general rule digital filters are not necessary.

In this chapter you will find explanations on filters for bandwidth limitation in SECTION 9.1 and on filters for integration and differentiation in SECTION 9.2.

9.1 Filters to Limit the Bandwidth

Without using digital filters, the signal contains frequencies between 0 Hz and half of the sample frequency. See CHAPTER 4 on this. This is optimal for evaluating spectra. As the FFT divides the signal up into the individual frequencies, the different frequencies do not disturb each other. In the time domain, signal portions of all frequencies are visible simultaneously. This means that the signal to be evaluated can under certain circumstances be covered by other signals. Digital filters are used to suppress these undesired signal portions.

In this section, you will find the following information:

- Operating principle - See SECTION 9.1.1 on this.
- Filter parameters - See SECTION 9.1.2 on this.

9.1.1 Operating Principle

Zero phase FIR filters are used. These filters only change the amplitudes of the spectral components, the phases remain unchanged.

The following example demonstrates the operating principle of the filters:

FIR filter, length 3

1/4	1/2	1/4

Low Frequency Signal (Approximately Sinusoidal)

–4	0	4	7	9	10	9	7	4	0	–4

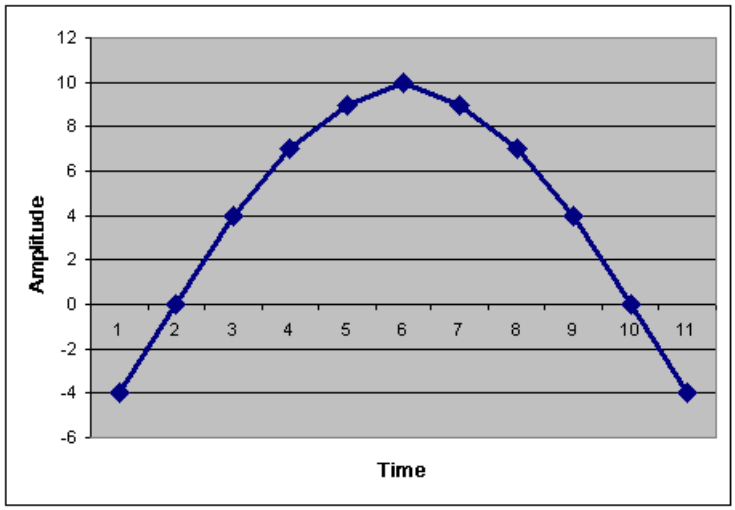


Figure 9.1: Graphic representation of a signal with a low frequency

To calculate the output signal, respectively three neighboring data points of the input signal are multiplied with the filter coefficients and the results are added together.

1st data point:

–4	0	4	7	9	10	9	7	4	0	–4
*	*	*								
1/4	1/2	1/4								
–1	0	1								
	+									
	0									

2nd data point:

–4	0	4	7	9	10	9	7	4	0	–4
	*	*	*							
	1/4	1/2	1/4							
	0	2	1.75							
		+								
	0	3.75								

3rd data point:

–4	0	4	7	9	10	9	7	4	0	–4
		*	*	*						
		1/4	1/2	1/4						
		1	3.5	2.25						
		+								
		0	3.75	6.75						

4th data point:

–4	0	4	7	9	10	9	7	4	0	–4
			*	*	*					
				1/4	1/2	1/4				
				1.75	4.5	2.5				
			+							
			3.75	6.75	8.75					

…

Result:

	0	3.75	6.75	8.75	9.5	8.75	6.75	3.75	0	

The low-frequency input signal is emitted nearly unchanged. The maximum deviation to the input signal is 5%.

High Frequency Signal

–8	0	8	–8	0	8	–8	0	8	–8	0

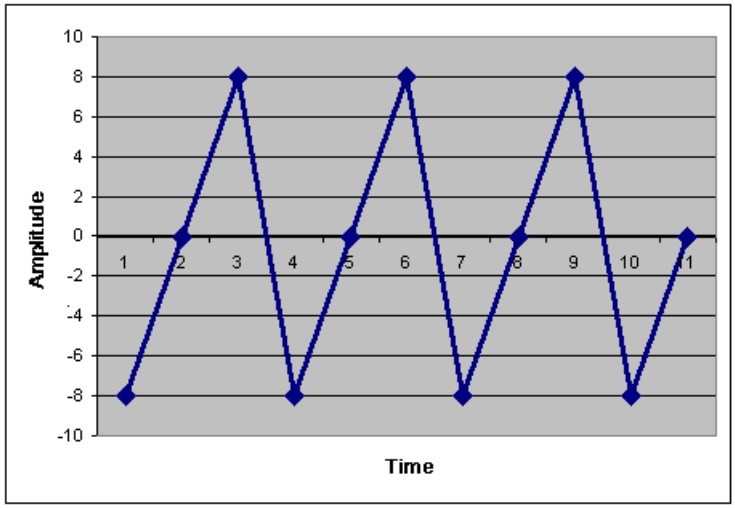


Figure 9.2: Graphic representation of a signal with a high frequency

1st data point:

–8	0	8	–8	0	8	–8	0	8	–8	0
*	*	*								
1/4	1/2	1/4								
–2	0	2								
+										
	0									

2nd data point:

–8	0	8	–8	0	8	–8	0	8	–8	0
	*	*	*							
	1/4	1/2	1/4							
	0	4	–2							
		+								
	0	2								

3rd data point:

–8	0	8	–8	0	–8	0	8	–8	0	8
		*	*	*						
		1/4	1/2	1/4						
		2	–4	0						
			+							
		2	–2							

…

Result:

	0	2	–2	0	2	–2	0	2	–2	

The filter attenuates the high-frequency signal in this example by a factor of 4.

As a conclusion, this example is a low pass filter which allows low frequencies to pass through and attenuates high frequencies. The example is just for explanation purposes, in the software longer filters are used. The longer the filter, the steeper the edges in the filter characteristic function.

The examples also make the following properties of the selected filtering clear:

- The input and output signal are not shifted against each other in time, i.e. the phase is not affected.
- The output signal is shorter than the input signal by the filter length –1.

For the output signal to have the desired length, the input signal has to be extended by the filter length –1. This is done automatically by the software. The extended input signal can not be detected by the user, apart from a slightly longer measurement time. When using the trigger, the trigger time is adjusted so that it applies to the filtered signal. When using several channels with different filter settings, the software takes care that the time signals are synchronous.

Filter characteristic

The properties of the digital filters are shown in a filter characteristic diagram. Attenuation is plotted over the frequency. Attenuation of 0 dB does not change the input signal. The range with 0 dB attenuation is called the pass band. In the stop band, the input signal is attenuated by the stop band attenuation. Between the pass band and the stop band there is the transition band. The cutoff frequency selected by the user is approximately in the middle of the transition band.

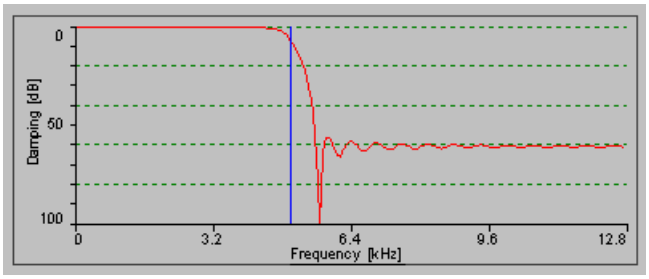


Figure 9.3: Filter characteristic

9.1.2 Filter Parameters

The following filter types are available:

- Low pass
- High pass
- Band-pass
- Band stop

The individual types are described below.

The Quality parameter affects both the length of the filters and also the approximate attenuation in the stop band. For illustration of the filter length the example filter [¼ | ½ | ¼] described above has a filter length of 3.

Quality	Length of filters	Stop band attenuation
very low	21	30 dB
low	41	40 dB
medium	61	50 dB
high	81	60 dB
very high	101	70 dB

Low pass Low pass filters allow low frequencies to pass and attenuate high frequencies.

High pass High pass filters allow high frequencies to pass and attenuate low frequencies.

Band-pass Band-pass filters allow medium frequencies to pass and attenuate high and low frequencies.

Band stop Band stop filters allow low and high frequencies to pass and attenuate medium frequencies.

9.2 Filters for Integration or Differentiation

Filters for integration and differentiation are used to convert the input signals displacement, velocity and acceleration into each other. The same applies to angle, angular velocity and angular acceleration.

ⓘ When evaluating the spectrum, it is possible to convert the signal during analysis.

With a time data evaluation, the signal already needs to be converted during the measurement. Converting the signal in the frequency and time domain can show different effects.

You will find the operating principle and the properties in the following sections:

- Filters for integration - See SECTION 9.2.1 on this.
- Filters for differentiation - See SECTION 9.2.2 on this.

9.2.1 Integration

Integration is used to convert velocity into displacement or acceleration into velocity. Integrating an acceleration signal twice to generate a displacement signal is not possible.

Operating principle

Integration is explained here using a velocity signal as an example which is converted into a displacement signal through integration. At the start of the time domain, it is arbitrarily defined that the displacement is to be zero. Therefore the first value of the displacement signal is 0. The second value of the displacement signal is determined by multiplying the first velocity value with the sample interval. The sample interval is the reciprocal of the sample frequency (sample interval = 1/sample frequency). The third value of the displacement signal is determined by multiplying the second velocity value with the sample interval and then adding the result to the second value of the displacement signal etc. The process is explained with the aid of the following figures.

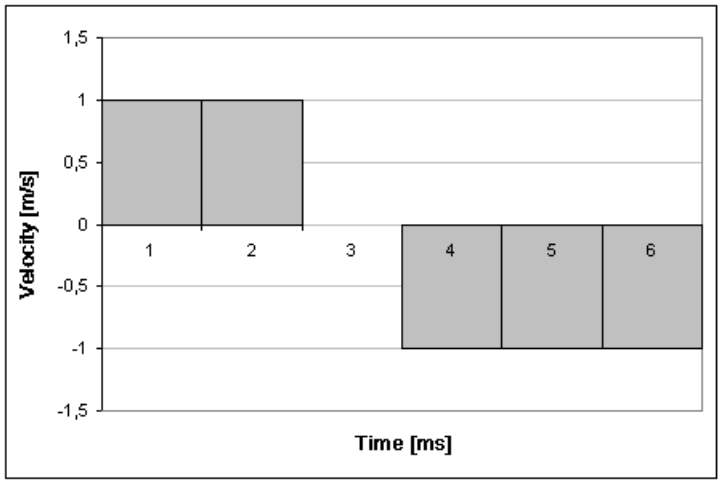


Figure 9.4: Measured velocity signal

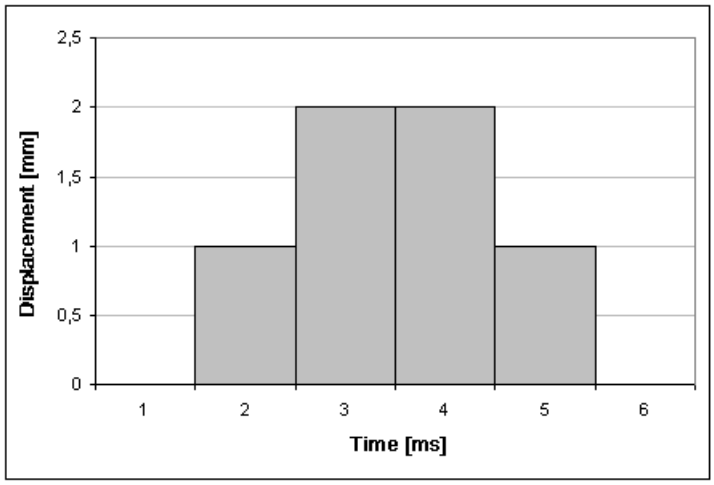


Figure 9.5: Calculated displacement signal

The displacement signal is generated, based on the assumption that the current velocity value remains unchanged until the next velocity value. In the case of constant velocity per section, it can be said that:

$$s_i = s_{i-1} + v \cdot \Delta t \qquad \text{Equation 9.1}$$

This assumption only leads to correct results with frequencies which are small in comparison to the sample frequency. In the case of frequencies which are nearly half the sample frequency, the approximation of a constant velocity per section can no longer be made. It would lead to errors of up to 30%. These errors are compensated for by a special FIR filter. See SECTION 9.1 on this. For frequencies up to 40% of the sample frequency (equivalent to 80% of the bandwidth), the resulting error is less than 0.5%.

Properties

Integration in the time domain usually only works satisfactorily if the time signal does not have a DC offset. An offset causes a ramp in the integrated signal. This ramp is often undesired, in particular for low sample frequencies.

The offset in the time domain can be suppressed in many cases by inserting a high pass filter. However, it must be ensured that this does not cut off any portions of the wanted signal.

During integration, a time delay of half a sample interval occurs (sample interval = 1/sample frequency). This leads to a frequency-dependent phase shift.

ⓘ If the time signal in the time window is periodic and the signal does not have a DC offset, then the integration in the time domain corresponds to the integration in the frequency domain. Otherwise the two results do not match exactly.

An integrated time signal as a general rule shows a DC offset because the first time value is arbitrarily set to zero. Because integrating a time signal with DC offset does not give any meaningful results, the software does not offer double integration.

9.2.2 Differentiation

Differentiation is used to convert displacement into velocity or velocity into acceleration. By differentiating twice, displacement can be converted into acceleration.

Operating principle

Differentiation is explained here using a displacement signal as an example which is converted into a velocity signal through differentiation. From two respective neighboring values of the displacement signal, the average velocity is determined by dividing the difference between the displacement values by the sample interval. The sample interval is the reciprocal of the sample frequency (sample interval = 1/sample frequency). The process is explained by the aid of the following figures.

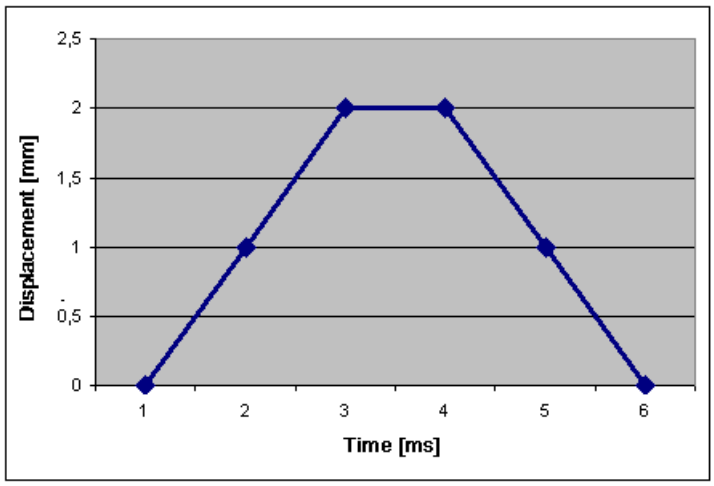


Figure 9.6: Measured displacement signal

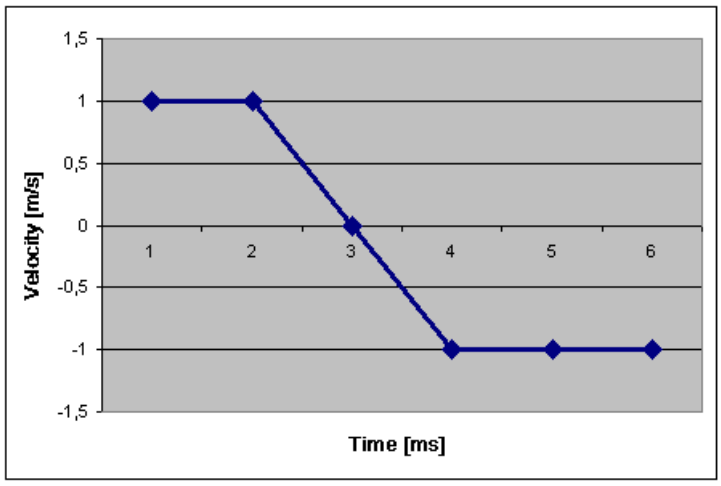


Figure 9.7: Calculated velocity signal

Thus it is presumed that the velocity between two neighboring samples is constant. In this case, it can be said:

$$v = (s_i - s_{i-1}) / (\Delta t) \qquad \text{Equation 9.2}$$

This assumption leads to correct results only with frequencies which are small in comparison to the sample frequency. In the case of frequencies which are nearly half the sample frequency, the approximation of a constant velocity per section can no longer be made. It would lead to errors of up to 30%. These errors are compensated for by a special FIR filter. See SECTION 9.1 on this. For frequencies up to 40% of the sample frequency (equivalent to 80% of the bandwidth), the resulting error is less than 0.5%. When differentiating twice, a maximum error of 1% can occur.

Properties

Differentiation in the time domain usually does not present any difficulties. Only a time delay of half a sample interval occurs during differentiation (sample interval = 1/sample frequency). This leads to a frequency-dependent phase shift.

ⓘ In the case of double differentiation there is no time delay. If the time signal is periodic in the time window, then the differentiation in the time domain corresponds to the differentiation in the frequency domain. Otherwise the two results do not match exactly.

10 VideoTriangulation

In this chapter you will find information on the principle of VideoTriangulation. See SECTION 10.1 on this. You will find how to do VideoTriangulation through the Modify 3D Coordinates dialog in SECTION 10.2 and through VideoTriangulation during Scan in SECTION 10.3. If you want to do VideoTriangulation for imported scan points, see SECTION 10.4. Useful quality criteria are described in SECTION 10.5. You will find hints on error handling in SECTION 10.6. If you want to do VideoTriangulation with a PSV-1D, see SECTION 10.7.

10.1 Principle of VideoTriangulation

VideoTriangulation is a software option (PSV-S-TRIA) which uses image processing to ensure that the laser(s) is/are positioned exactly on the scan point before the measurement. With the PSV-3D, VideoTriangulation has an additional task, measuring the 3D coordinates of the scan points.

VideoTriangulation has the following tasks:

- Compensating for errors in the 3D coordinates of the scan points
- Compensating for deviations between the actual scanners and their mathematical description
- Compensating for temperature drift in the scanners
- Compensating for errors in the 3D alignment

The key component for VideoTriangulation is detecting the laser beam in the video image. Using the image processing algorithms implemented, the position can be determined with sub-pixel accuracy. To obtain optimal measurement results, the contrast of the laser beam in the video image is optimized. For this reason the video image is mostly black during VideoTriangulation and only the high intensity laser beam can be seen on it.

Another component of VideoTriangulation is a control algorithm to move the laser beam to a new position in the video image. This control algorithm uses 2D alignment to calculate the adjustment of the laser. For this reason a valid 2D alignment must be available. With objects that are not flat, the 2D alignment can not be perfectly correct in all positions. To make sure that the control algorithm does not oversteer in these cases, the control is carried out in several steps. In every step, the laser beam is only moved around half the theoretically calculated distance. As the remaining deviation halves respectively, the laser beam reaches the target location in a few steps.

For the PSV-3D, a new 3D coordinate is calculated from the updated laser angles once all three laser beams have been moved to the target position. This is the reason for the Triangulation component in the name VideoTriangulation. As a general rule, the laser beams do not meet exactly, but form skew lines. This is why for the new 3D coordinates, the point is taken for which the straight lines come closest to each other.

Once VideoTriangulation has been carried out on one point, a calculation is carried out to work out by how much the angles of the laser beams needs to be corrected so that they meet exactly at the calculated 3D coordinate. With all the following positioning processes for 3D coordinates, the laser angles are corrected by these calculated angles (pre-correction). This mechanism

allows the laser beams to meet at the next scan point relatively well, even before video triangulation, providing the 3D coordinates are correct. In particular at the edges of the object, this mechanism has the effect that a laser beam does not fall off the object which would make it impossible to detect in the video image and therefore make VideoTriangulation impossible.

10.2 VideoTriangulation using the Modify 3D Coordinates Dialog

In the Modify 3D Coordinates dialog, click VideoTriangulation to carry out VideoTriangulation for the point currently selected and to recalculate the coordinates. If you apply the coordinates with OK, the scan point is given the Video Triangulated Geometry Status. If you modify the coordinates after VideoTriangulation, then the scan point is given the Modified Geometry Status.

If several scan points had been selected when calling up the dialog, VideoTriangulation is only carried out for one of these scan points. The coordinates of the other scan point are shifted by the same vector as the scan point for which VideoTriangulation was carried out. The coordinates of the other scan points are given the Modified Geometry Status.

10.3 VideoTriangulation during Scan

The following modes are available for VideoTriangulation:

- Standard
- Fast
- Off (only PSV-3D)

Standard

The following criteria apply to the Standard mode:

- VideoTriangulation is carried out on every scan point during the scan before the vibration measurement data is captured.
- After VideoTriangulation, the 3D coordinates and if applicable also the 2D coordinates are updated.
- If VideoTriangulation is successful, the scan points are given the Video Triangulated Geometry Status.
- If VideoTriangulation fails, the Geometry Status remains unchanged and no vibration measurement data is captured, the scan points are given the VT Failed Scan Status.

Fast

The following criteria apply to the Fast mode:

- During a scan, VideoTriangulation is carried out on the first scan points.
- Once VideoTriangulation has been successful carried out on 3 scan points, the angular deviation ascertained is applied to the next scan point without checking whether the laser beams meet.
- For every 10th scan point, a VideoTriangulation is carried out to update the angular deviation.
- If the lasers are moved to another image section of the video image (distance in x- or y-direction more than 200 pixels), the fast VideoTriangulation is reinitialized.
- Fast VideoTriangulation can only intersect the lasers on a point if correct 3D coordinates are available for all scan points. For this reason, we only recommend using fast VideoTriangulation if all scan points have the Video Triangulated Geometry Status.

ⓘ If VideoTriangulation Standard has failed for several points, then it is possible to try to move the lasers to the target position using the Modify 3D Coordinates dialog. If this is not successful, these scan points should be disabled before starting the scan.

Off (only PSV-3D)

The following criteria apply to the Off mode:

- If VideoTriangulation Standard was carried out previously, the pre-correction determined for the scanner angles is applied.

  This can remove any skewness of the laser beams which may exist.
- If the 3D coordinates are correct, then the pre-correction considerably reduces the distance between the laser beams on the object.
- To determine the angles of the pre-correction, it is enough to carry out VideoTriangulation on one scan point. This can also be done using Modify 3D Coordinates. In this case it doesn't matter if the modified coordinates are applied with OK or not.
- The values for the pre-correction are deleted if the 3D alignment is carried out again. For this it is sufficient to open the 3D Alignment dialog and to click Calculate.

10.4 VideoTriangulation for Imported Scan Points

In several cases the 2D coordinates of the scan points are calculated from the 3D coordinates. This is the case with the following:

- Imported 3D geometry
- Changing the alignment and confirming the following message with Yes: The alignment has changed. Do you want to recalculate the positions of the scan points on the video image?

In these cases, the 2D coordinates of the scan points are calculated from the 3D coordinates. The calculated 2D coordinates are approximate values and therefore do not lie exactly on the point that the laser is positioned at on the video image. For this reason, the calculated 2D coordinates are generally not used for VideoTriangulation.

For all scan points where the 2D coordinates have been calculated from the 3D coordinates, at the start of the VideoTriangulation, first of all the initial positions of the laser beams in the video image are determined. The actual VideoTriangulation is only carried out after this.

ⓘ With the PSV-1D, the 2D coordinates are set at the video coordinates found for the laser. The 3D coordinates remain unchanged.

With the PSV-3D, the 2D coordinates are set to the center point of the video coordinates found for the three lasers. The lasers are positioned to these 2D coordinates and the 3D coordinates are updated.

10.5 Quality Criteria

The following quality criteria are important for successful VideoTriangulation:

- It must be possible to position all three lasers to at least 2 pixel accuracy on the target point in the video image.
- The distance between the old and the new 3D coordinates must be smaller than the value set. You set this distance in the Preferences dialog on the Geometry page. To do so, select Setup > Preferences and display the Geometry page.

ⓘ This distance criterion is ignored for scan points with the Triangulated Geometry Status, for VideoTriangulation in the Modify 3D Coordinates dialog and for VideoTriangulation in Fast mode.

10.6 Error Handling

The following errors are recorded in the message box of the PSV software:

- VideoTriangulation failed because it was not possible to position the lasers to within 2 pixel accuracy on the target coordinates.
- VideoTriangulation failed because a laser could not be found.
- An angle calculated during pre-correction exceeds the threshold value of 0.3°. Consequently: Pre-correction is not active, i.e. for positioning the laser beams the uncorrected angles are used.

10.7 VideoTriangulation for PSV-1D

With the PSV-1D, VideoTriangulation is also permitted (requires PSV-S-TRIA option). With the PSV-1D, the only function of VideoTriangulation is to position the laser beam precisely on the required scan point. A real measurement of the 3D coordinates is not carried out.

Depending on the origin of the 2D coordinates, the following happens:

- 2D coordinates were calculated from 3D coordinates:

  The 2D coordinates are determined by identifying the laser beam on the video image, the 3D coordinates remain unchanged and the Geometry Status remains unchanged.
- 2D coordinates were not calculated from 3D coordinates:

  The laser is positioned at the 2D coordinates, the 3D coordinates are recalculated using the same distance and the updated scanner angles and the Geometry Status becomes Video Triangulated.

With the PSV-1D, VideoTriangulation can only be used with active 3D alignment and available 3D coordinates.

11 Working Principle of the Bandwidth Extension for Heterodyne Carrier Signals (UHF)

In this chapter you can find the theoretical background of the bandwidth extension being available for the UHF as described in SECTION 11.1, and why it is useful to expand the bandwidth mathematically as described in SECTION 11.2. In SECTION 11.3 you can see how lower order side bands are calculated, and SECTION 11.4 contains a simplified version of the algorithm that is used by the software to expand the bandwidth.

11.1 Introduction

Frequency modulation (fm) theory states that heterodyne carrier signals cannot be frequency modulated with modulation frequencies f_m higher than the carrier frequency f_c subtracted by the maximum frequency deviation $f_{d,max}$. Thus, $f_m \leq f_c - f_{d,max}$ is usually considered as maximum limit to obtain the correct modulation signal from the carrier signal by fm demodulation. This limit can be breached for small phase modulation indices with the technique discussed here. A special algorithm makes possible to demodulate heterodyne carrier signals with a modulation frequency of up to twice the carrier frequency if a small phase modulation index M can be assumed. This assumption is valid in heterodyne laser interferometers for ultrasonic testing where the phase modulation of a detector signal is originated by vibration amplitudes much smaller than the wavelength of the laser.

The core idea of the bandwidth extension procedure is as follows: the components of the spectrum of the phase-modulated carrier at frequencies lower the carrier frequency have a well-defined phase relation to the components at the higher frequencies if small modulation indices can be assumed. The lower frequency components can be computed in the case of a digitized carrier signal with a unitary operation from the higher frequency components. Therefore, only the carrier and the components higher than the carrier frequency have sufficient information to obtain the phase modulation. A demodulation-frequency bandwidth of twice the carrier frequency is possible and, thus, an algorithm to demodulate a digitized carrier signal with a bandwidth of twice the carrier frequency is employed to expand the bandwidth of the UHF from 600 MHz to 1.2 GHz.

11.2 Why Expanding the Bandwidth Mathematically?

The optical sensor of the UHF is the Mach-Zehnder interferometer shown in FIGURE 11.1. The photo diode of the heterodyne interferometer creates a current:

$$i_S(t) = K(P_m + P_r + 2\varepsilon\sqrt{P_m + P_r}\cos(\omega_c t - \varphi(t))) \qquad \text{Equation 11.1}$$

ω_c... Angular carrier frequency

The current amplitude is dependent on the conversion parameter of the detector K = ηq/hv, the power of the measurement light impinging the detector P_m, and the power of the reference light P_r. q = 1.602177 · 10^{-19} C is the charge of an electron, h = 6.6261 · 10^{-34} J is the Planck's constant, η is the quantum efficiency of the detector, and v = c/λ is the laser-light frequency. The speed of light is c = 299792458 m/s and λ is the wavelength. In addition, ε is the heterodyning efficiency, which takes into account any degradation of the photocurrent due to optical distortions or misalignments, and the time dependent phase modulation by the specimen displacement s(t) is defined as:

$$\varphi(t) = \frac{4\pi}{\lambda}s(t) \qquad \text{Equation 11.2}$$

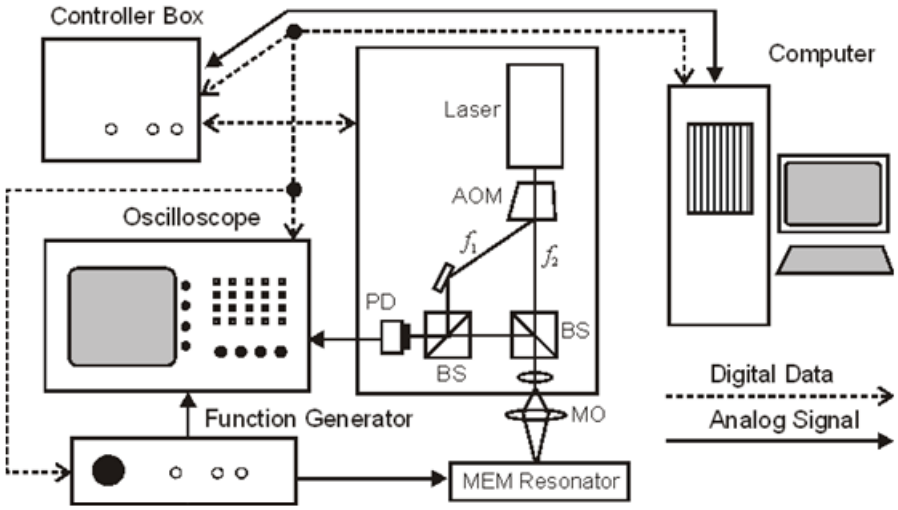


Figure 11.1: Schematic of the interferometer system. BS is a beam splitter, PD is a photo detector, MO is a microscope objective, and AOM is the acousto-optical modulator (Bragg cell).

The Bragg cell in the interferometer causes a frequency shift of approximately 615 MHz between reference and measurement beams. Thus, the photo detector with a flat amplitude-transfer characteristic detects a heterodyne carrier signal which is phase-modulated by the vibrations of the specimen.

The RF detector voltage signal $u_s(t) = K_u i_s(t)$ is obtained from a trans-impedance amplifier with amplification K_u and is sampled with a digital high-frequency oscilloscope as see in FIGURE 11.1. The data is transferred to a PC via a local area network (LAN) connection. Thus, the digitized photo-detector signal can be processed completely in a PC. Here, the signal processing of the detector signal is performed numerically.

The generation of a constant interferometric carrier signal at a high frequency for a low noise, high-power laser source is not a simple task and is associated with technical effort and cost. Therefore, mathematics is applied to obtain a bandwidth of twice the carrier signal.

11.3 Computations of Lower Order Side Bands

Every frequency component of the vibration spectrum generates an independent sinusoidal phase modulation of the carrier. Thus, every frequency component can be considered independently from the amplitudes of the remaining spectrum. The heterodyne carrier signal with a single frequency, phase modulation $\varphi(t) = (4\pi/\lambda)\hat{s}\cos(\omega_m t + \varphi_m)$ of a heterodyne interferometer can be written in general by:

$$u_s(t) = U_S \sin(\omega_c t + \varphi_c + (4\pi/\lambda)\hat{s}\cos(\omega_m t + \varphi_m)) \qquad \text{Equation 11.3}$$

Or by:

$$u_S(t) = U_S \sin(\omega_c t + \varphi_c + M\cos(\omega_m t + \varphi_m)) \qquad \text{Equation 11.4}$$

$M = 4\pi\hat{s}/\lambda$ … Modulation index

U_S … Amplitude

$\hat{s}$ … Vibration amplitude

ω_c … Carrier angular frequency

φ_c … Carrier phase

ω_m … Modulation angular frequency

φ_m … Modulation phase

Equation (11.1) can be transformed into:

$$u_S(t) = U_S(\sin(\omega_c t + \varphi_c)\cos(M\cos(\omega_m t + \varphi_m)) + \cos(\omega_c t + \varphi_c)\sin(M\cos(\omega_m t + \varphi_m))) \qquad \text{Equation 11.5}$$

The Jacobi-Anger expansion

$$e^{iz\cos\theta} = J_0(z) + 2\sum_{n=-\infty}^{\infty} i^n J_n(z) e^{in\theta} \qquad \text{Equation 11.6}$$

is employed to rewrite Equation (11.5) into

$$\begin{aligned} u_S(t) &= U_S \sin(\omega_c t + \varphi_c) \\ &\left(J_0(M) + 2\sum_{n=1}^{\infty} (-1)^n J_{2n}(M) \cos(2n(\omega_m t + \varphi_m)) \right) \\ &+ U_S \cos(\omega_c t + \varphi_c) \\ &\left(-2\sum_{n=1}^{\infty} (-1)^n J_{2n-1}(M) \cos((2n-1)(\omega_m t + \varphi_m)) \right) \end{aligned}$$

Equation 11.7

$J_n(M)$… nth order Bessel function

Assuming a small modulation index M <<1 equation (11.7) can be approximated by the following equation because the higher order Bessel terms are small compared to the zero and first order term.

$$\begin{aligned} u_S(t) &= U_S[J_0(M)\sin(\omega_c t + \varphi_c) \\ &+ 2J_1(M)\cos(\omega_c t + \varphi_c)\cos(\omega_m t + \varphi_m)] \end{aligned}$$

Equation 11.8

At next the relation

$$\begin{aligned} \cos(\omega_c t + \varphi_c)\cos(\omega_m t + \varphi_m) &= \frac{1}{2}\cos((\omega_c - \omega_m)t + \varphi_c - \varphi_m) \\ &+ \frac{1}{2}\cos((\omega_c - \omega_m)t + \varphi_c - \varphi_m) \end{aligned}$$

Equation 11.9

and the relation $J_{-n}(z) = (-1)^n J_n(z)$ yield

$$\begin{aligned} u_S(t) &= U_S J_0(M)\sin(\omega_c t + \varphi_c) \\ &- U_S J_{-1}(M)\cos((\omega_c - \omega_m)t + \varphi_c - \varphi_m) \\ &+ U_S J_1(M)\cos((\omega_c + \omega_m)t + \varphi_c + \varphi_m) \end{aligned}$$

Equation 11.10

In the following it is assumed that the periods of all three frequency components ω_c, $\omega_c - \omega_m$, and $\omega_c + \omega_m$ would fit in an time interval as integer. Then the coefficients of the Fourier series extension can be computed with $\hat{J}_0 = |\hat{J}_0|(a_0 + ib_0)$, $\hat{J}_{-1} = |\hat{J}_{-1}|(a_{-1} + ib_{-1})$, and $\hat{J}_1 = |\hat{J}_1|(a_1 + ib_1)$.

The coefficients for the 0 order Bessel line $a_0 + ib_0$, the –1 order Bessel line $a_{-1} + ib_{-1}$, and the +1 order Bessel line $a_1 + ib_1$ can be computed as coefficients of a Fourier series. For example a_1 yields from

$$\begin{aligned} a_1 &= \frac{1}{\pi} \int_{-\pi/\omega_c + \omega_m}^{\pi/\omega_c + \omega_m} \cos((\omega_c + \omega_m)t + \varphi_c + \varphi_m) \\ &\quad \cos((\omega_c + \omega_m)t)(\omega_c + \omega_m)dt \\ &= \cos(\varphi_c + \varphi_m) \end{aligned}$$

Equation 11.11

The complex coefficients a_0, b_0, b_1, a_{-1}, and b_{-1} are derived in the same way and result to $a_{-1} + ib_{-1} = \cos(\varphi_c - \varphi_m) - i\sin(\varphi_c - \varphi_m)$, $a_0 + ib_0 = \sin\varphi_c + i\cos\varphi_c$, and $a_1 + ib_1 = \cos(\varphi_c + \varphi_m) - i\sin(\varphi_c + \varphi_m)$. Thus the following relations are valid:

$$\begin{aligned} a_0 &= \sin\varphi_c \\ b_0 &= \cos\varphi_c \\ a_{-1} &= -\cos\varphi_c\cos\varphi_m - \sin\varphi_c\sin\varphi_m \\ b_{-1} &= \sin\varphi_c\cos\varphi_m - \cos\varphi_c\sin\varphi_m \\ a_1 &= \cos\varphi_c\cos\varphi_m - \sin\varphi_c\sin\varphi_m \\ b_1 &= -\sin\varphi_c\cos\varphi_m - \cos\varphi_c\sin\varphi_m \end{aligned} \qquad \text{Equation 11.12}$$

Employing some arithmetic equation (11.12) yields

$$\begin{aligned} a_{-1} &= (a_0^2 - b_0^2)a_1 + 2a_0b_0b_1 \\ b_{-1} &= 2a_0b_0a_1 - (a_0^2 - b_0^2)b_1 \end{aligned} \qquad \text{Equation 11.13}$$

Therefore, the $\hat{J}_{-1}$ side band can be computed by the final equation:

$$\hat{J}_{-1} = -|\hat{J}_1|[(a_0^2 - b_0^2)a_1 + 2a_0b_0b_1 + i(2a_0b_0a_1 - (a_0^2 - b_0^2)b_1)]\ldots \qquad \text{Equation 11.14}$$

For small modulation indices of a heterodyne carrier signal the J_0 and J_1 components contain all the information and J_{-1} can be derived. Therefore, the carrier and the right-hand side of the spectrum of the heterodyne carrier signal in respect to carrier contain all the information and are sufficient for the demodulation. Consequently, $f_m < f_c - f_{d,max}$ is not the ultimate limit for the modulation frequency of a heterodyne carrier signal with a phase modulation index M << 1. It is just necessary to evaluate the carrier and the right-hand side of the spectrum of the heterodyne carrier signal in respect to carrier to obtain the limit $f_m < 2f_c$. In the next section an algorithm is presented to evaluate just this part of a spectrum of a heterodyne carrier signal to expand the demodulation bandwidth.

11.4 Simplified Working Principle of the Algorithm to Expand the Bandwidth

The analog digital converter of the oscilloscope samples the heterodyne carrier signal from the detector with a sample frequency f_S during the sample time T_S. This results in $n = |f_S \cdot T_S + 1|$ samples $A[i_n]$ where n is an integer and $i_n = 1, 2 \ldots n$. The basic idea of the algorithm to expand the bandwidth of the discrete Fourier transform (DFT) spectrum of $A[i_n]$ is quite simple. The entry of the carrier and the frequency components higher than the carrier are shifted to higher frequency entries of the discrete spectrum. The entries lower than then carrier frequency are deleted. At next, the entries in the expanded spectrum, lower than the shifted carrier frequency, are computed with equation (11.13). The concrete structure of the DFT makes it necessary to distinguish an even and odd number of time samples and, thus the computation becomes more complex.

The first entry $F[i_n = 0]$ of the complex spectrum $F[i_n]$ obtained from a DFT is a signal offset at frequency zero. The remaining entries of the complex spectrum $F[i_H] = F[i_{L,max} – i_L + 1]$ for even n and $F[i_H] = F[i_{L,max} – i_L + 2]$ for odd n are flipped where i_L are the indices $i_L = 2 \ldots (n + 1)/2$ for odd n and $i_L = 2 \ldots n/2 + 1$ for even n. This context is illustrated in FIGURE 11.2. The highest index of i_L is $i_{L,max} \cdot i_H = i_{L,max} + 1 \ldots i_{H,max}$ are all indices of i which are not element of i_L and $i_{H,max}$ is the highest index of i_H. i_L and i_H have the same number of elements. $F[i_H] = F[i_{L,max} – i_L + 2]$ means $F[i_H(j)] = [i_{L,max} – i_L(j) + 2]$ for $1, 2 \ldots i_{L,max}$. The highest frequency entry occurs just one time in case of an even n. The offset entry occurs just one time for even and odd n. The single frequency entries are separated by $\Delta F = F[i_n + 1] – [i_n] = 1/T_S$. At next the expanded spectrum F_{exp} with 2m additional entries is computed ($m \in N$).

This starts with the determination of the index i_C of the carrier frequency f_C in the complex spectrum: $f_C = F[i_C]$ and is demonstrated in FIGURE 11.2, a). At next the entries with an index from i_L are shifted about m digits to higher indices. Thus, $f_c = F[i_n] \Rightarrow f_{c,exp} = f_c + m\Delta F = F[i_{c,exp} = i_c + m]$ for the carrier frequency and $F_{exp}[i_n + m] = F[i_n]$ for the indices $i_n > i_c$. The resulting expanded spectrum F_{exp} has m more entries for i_L which are called $i_{L,exp}$ in the following. The entries of the expanded spectrum at indices smaller $i_{c,exp}$ are derived from the entries $F[i_n]$ with an index higher i_c which correspond to the entries $F_{exp}[j]$ with an index higher $i_{c,exp}$.

The entries smaller $i_{c,exp}$ are deleted as seen in FIGURE 11.2, b) and computed by employing equation (11.13) as shown in FIGURE 11.2, c). Therefore, a spectrum with a mΔF shifted carrier frequency is constructed to avoid influences of the frequency components at the frequencies smaller the carrier frequency in the photo-detector signal. Such frequency components would falsify the measurements and are eliminated with the presented method.

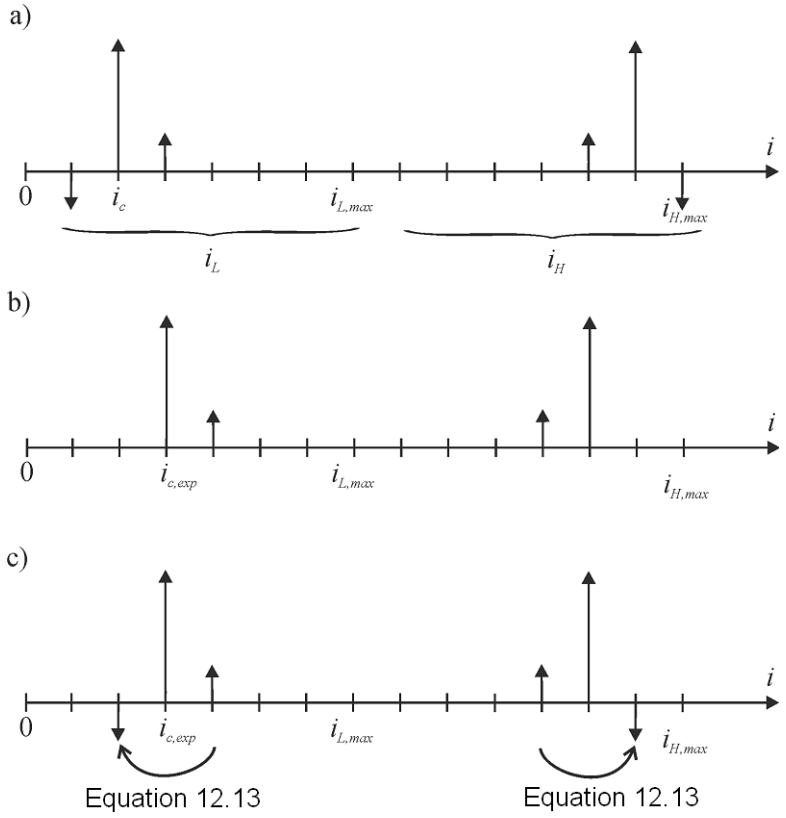


Figure 11.2: Frequency axis of a DFT spectrum with the indices i_c, $i_{L,max}$, and $i_{H,max}$

The sets for i_L and i_H are also shown in FIGURE 11.2. Graphic a) shows the original spectrum, b) shows the shifted components, and c) illustrates the computation of the deleted low-frequency components with equation (11.13).

The number of entries in the $i_{L,max}$ has to be at least $2i_C + m$ before the bandwidth extension to ensure unambiguousness. The sample rate has to be high enough to ensure $i_{L,max} \geq 2i_C + m - 1$ if $n + 2m$ is an odd number and $i_{L,max} \geq 2i_C + m - 2$ if $n + 2m$ is an even number. The sample rate has to be at least 6 times the carrier frequency if the bandwidth is doubled. For example, the sample rate has to be > 3.6 GHz if the bandwidth is extended from 600 MHz to 1.2 GHz to meet the Shannon-Nyquist-Theorem.

Finally the signal with the expanded spectrum $F_{exp}[j] \cdot (j = 1, 2 \ldots n + m)$ of the detector signal is demodulated with the arctan method shown in FIGURE 11.3.

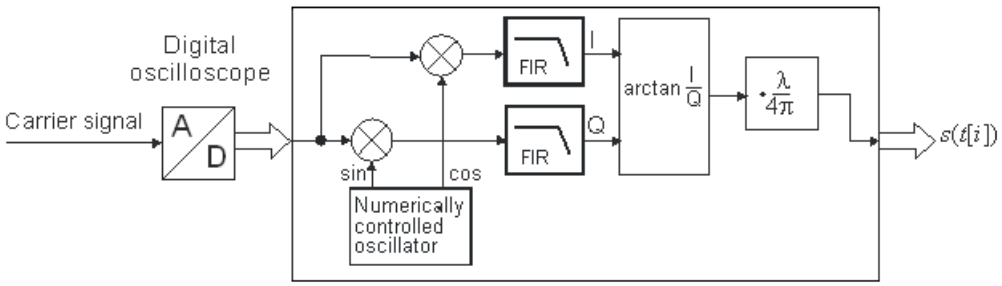


Figure 11.3: Scheme of the digital demodulation with the arctan method

The expanded carrier frequency $f_{c,exp} = f_c + m\Delta F$ is twice the original carrier frequency if the demodulation bandwidth is doubled. The expanded carrier frequency is used as sine signal to convert the heterodyne carrier signal down into the base band and by multiplying also with a 90° phase-shifted cosine signal an in-phase and a quadrature (IQ) signal is obtained after low-pass filtering the I and the Q signal. Finally the arctan2 operation is employed to compute the phase.

Leakage appears when complex spectra of real measurement signals are evaluated. Thus, more advanced algorithms as the one described here have been employed in the UHF system to save computation time and to minimize leakage effects.

12 Theoretical Background for PC-Based Digital Demodulation (VDD)

In the sensor head a heterodyne interferometer generates a high-frequency carrier signal with the aid of a Bragg cell. For the digital demodulation, the carrier signal is converted into two quadrature signals I and Q in a mixing process. The quadrature signal pair carries the same information as the carrier signal, however it lies in the base band and is thus easy to process electronically.

Ideally the quadrature signals I and Q are sinusoidal, have the same amplitudes and are phase-shifted 90° apart. In the following figure, we can see that in an I-Q diagram they depict a rotating pointer whose angle of rotation is equal to the interferometric phase difference Δφ. The sense of rotation corresponds to the direction of movement of the object under investigation.

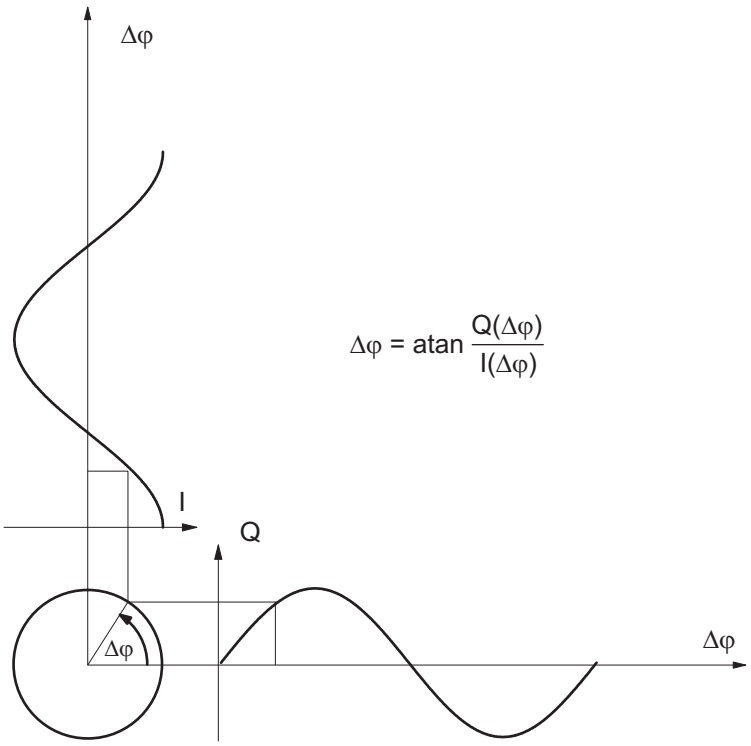


Figure 12.1: Graphical representation of the quadrature signals I and Q

The phase difference Δφ is calculated using the following trigonometric relationships.

$$I(\Delta\varphi) = A\cos\Delta\varphi \qquad \text{Equation 12.1}$$

$$Q(\Delta\varphi) = A\sin\Delta\varphi \qquad \text{Equation 12.2}$$

We divide equation (12.2) by equation (12.1) and obtain the following relation.

$$\frac{Q(\Delta\varphi)}{I(\Delta\varphi)} = \frac{\sin\Delta\varphi}{\cos\Delta\varphi} = \tan\Delta\varphi$$ Equation 12.3

After applying the inverse tangent function we finally get the following.

$$\Delta\varphi = \operatorname{atan}\frac{Q(\Delta\varphi)}{I(\Delta\varphi)}$$ Equation 12.4

The phase difference Δφ is proportional to the displacement Δx of the object according to the following relation.

$$\Delta\varphi = \frac{4\pi}{\lambda}\Delta x$$ Equation 12.5

In practice it is important to generate the quadrature signals with a high level of stability and spectral purity. Furthermore a suitable A/D converter is required to sample both the quadrature signals with sufficient resolution. After sampling, the remaining errors - which are mainly caused by the analog components in the mixing process - are corrected using special algorithms.

13 Working Efficiently with the PSV

In this chapter, you will find explanations on how the PSV records measurement data of a vibrating system in SECTION 13.1.

If you want to optimize the measurement setup, read SECTION 13.2.

Suitable settings for your measurement are described in SECTION 13.3.

You will find information on focusing the laser beam in SECTION 13.4.

In SECTION 13.5 you are shown how to reduce noise using the tracking filter.

To optimize the input signals, proceed as described in SECTION 13.6.

With overlap, you can reduce the time for the measurement as described in SECTION 13.7.

13.1 Recording Measurement Data in a Vibrating System

A typical measurement setup with the PSV is shown in the following figure.

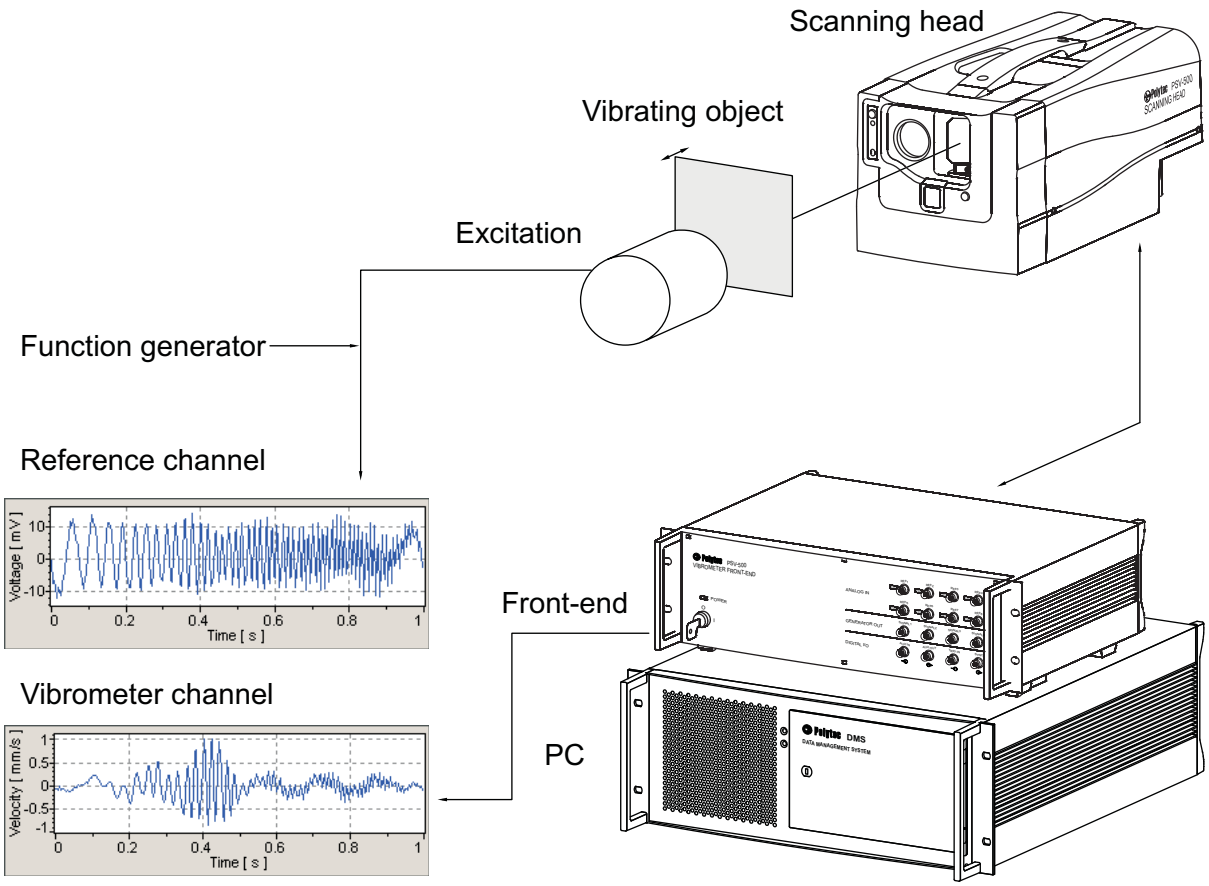


Figure 13.1: Principle of data recording with the PSV

The following steps are to be taken into account:

- An excitation (shaker, loudspeaker, hammer etc.) causes the object under investigation to vibrate.
- The measurement beam from the interferometer in the scanning head is positioned to a scan point on the object by means of mirrors and is scattered back.
- The backscattered laser light interferes with the reference beam in the scanning head.
- A photo detector records the interference.
- A decoder in the controller provides a voltage which is proportional to the velocity of the vibration parallel to the measurement beam.
- The voltage is digitized and processed as vibrometer signal.

It is important to select the kind of excitation which allows you to receive the best results for your measurement task. The type of excitation also decides on the measurement parameters. The following types of excitation are available:

- Periodic i.e. with a repeating signal (sinusoidal vibration, periodic chirp, pseudo random, etc.)
- Transient i.e. with a pulse (e.g. rectangular pulse, hammer blow etc.)
- Stochastic (random) i.e. with noise (e.g. noise generator, self-excited)

You can use the excitation as a reference signal or for triggering. You can also use the excitation force measured by a force cell as a reference signal. The reference signal is processed in the same way as the vibrometer signal.

13.2 Optimizing the Measurement Setup

Position the scanning head

The PSV measures velocity components parallel to the laser beam. Position the scanning head and the object in the following way:

- The laser beam can cover the complete area to be scanned.
- The front panel of the scanning head and the center of the area to be scanned are located opposite each other.
- The longitudinal axis of the scanning head is positioned perpendicular to the area to be scanned.

Optimize the signal level

Test the vibrometer signal level (in the scanning head control) at several scan points, especially at scan points with unfavorable backscattering behavior. The higher the level, the better the signal-to-noise ratio will be. The PSV often also provides useful measurement results at low levels.

1. If the vibrometer signal level is too low, check the focus of the laser beam. Use Assign Focus during Scan if one focus value does not apply for all scan points.
2. Check the distance between the front panel of the scanning head and the measured object.
3. Increase the distance if the laser beam hits the lateral points of the scan field at a great angle.
4. Try to place the object in a maximum of the laser intensity.

   The distances between the maxima are 204 mm (= length of the laser resonator). In between there are minima of the laser intensity where strong temporal fluctuations of the signal intensity can appear. You will find more detailed information on visibility maxima in your hardware manual.
5. If necessary, treat the surface of the area to be scanned.

   Highly reflective and transparent surfaces usually do not scatter enough laser light back to the scanning head. Evenly matt surfaces are ideal, as for example achieved by means of a sandblaster. A useful tip is to cover a large object with a coat of paint using water-soluble white wall paint or Spanish white. You can also use Ardrox® developer spray or reflective film.

ⓘ If your PSV is equipped with the CoherenceOptimizer®, you get good results at all working distances. Move your scanning head to the coherence maxima for optimal results.

13.3 Optimizing Resolution and Measurement Time

The total time needed for a scan is:

$$t_T = t_W \cdot n_p \cdot n_a + t_{add}$$ Equation 13.1

- t_T … Total time for a scan
- t_W … Length of the time window
- n_p … Number of scan points
- n_a … Number of averages
- t_{add} … Additional times

Additional times for a scan are, for example:

- Time to wait for a trigger
- Time for remeasurement
- Time to wait until the scanner mirrors are at rest
- Time to update measurement data in analyzers
- Computing time

You can optimize the resolution and the time needed for a scan as follows:

1. Select a sufficient number of FFT lines to obtain an appropriate number of frequency values in the set bandwidth.
2. Select the number of scan points according to the size and structure of your scan field.
3. Select the number of averages depending on the noise level.
4. Select Overlap if available. See SECTION 13.7 on this.
5. Avoid triggering if possible.
6. Close analyzers which you do not need.

13.4 Focusing the Laser Beam

The laser is focused using an integrated stepper motor.

There are several ways of adjusting the focus:

- Manual: Using the N and F keys in the scanning head control.
- Autofocus (complete): The complete focus range is traversed and the signal level is evaluated. Then the focus is set to the position at which the maximum signal level occurred.
- Autofocus (fast): The focus is traversed in a small range around the current position. Then the focus is set to the position at which the maximum signal level occurred.
- Assign Focus Automatically: Autofocus is carried out at several measurement points. From the focus values determined and the focal length of the lens, the infinite focus position is determined. This new infinite focus position is used to calculate the focus values for all scan points.
- Calculate Focus Values Automatically: The focus values for the scan points are calculated from the distances of the measurement points from the scanning head and are assigned to the scan points. The calculated focus values are set if Focusing during Scan is switched on and if either a scan point is clicked or the scan point is hit during the scan and the current focus position is more than the depth of focus away from the calculated position.

The following information is required to calculate the focus values:

- Focal length of the lens
- Infinite Focus Position of the motor for adjusting the lens
- Distance of the measurement point from the front panel of the scanning head

After performing Assign Focus Automatically, the software determines the Infinite Focus Position. With this infinite focus position, he focus values for all scan points are calculated.

The software offers to update the Infinite Focus Position in the Preferences dialog. This option is offered every time after Assign Focus Automatically has been selected, providing the values determined for the Infinite Focus Position are plausible. Even if the focal length entered has an error of several mm, this updated Infinite Focus Position allows the focus values to be calculated as long as the measurement distance does not vary much. Calculating the focus values however, depends on the 3D geometry being correct.

The nominal values for the focus calculation are:

	PSV-400	**PSV-500**
Infinite focus position	2850 steps	7500 steps
Focal length	LR: 98.5 mm MR: 60.0 mm	70 mm

13.5 Noise Reduction with the Tracking Filter

Principle

For weak optical signals (dark surfaces, long distances), it is important to select a suitable tracking filter setting (Acquisition Settings dialog, Vibrometer page). Due to the physical principle of the measurement (speckle effect), the back scattered laser light can intermit for a short time. The tracking filter bypasses these intermittences and decreases the noise of the input signal according to the PLL Principle (Phase Locked Loop).

If the tracking filter is activated, an oscillator is coupled to the input signal. The oscillator has the effect of a flywheel.

The oscillator follows the input signal if the following applies:

- Velocities and accelerations which are not too high appear on the scan field. The acceleration is proportional to the product of velocity and frequency of the measured vibration.
- The input signal intermits for a short period only.

The oscillator does not follow the input signal and the output signal is of no use if the following applies:

- Very high velocities and accelerations appear on the scan field.
- The input signal intermits for a long period.

Settings

You will find detailed information on selecting suitable tracking filter settings in your hardware manual.

ⓘ In the PSV-400, when using the digital VD-07 or VD-08 velocity decoder, the tracking filter is bypassed and does not have any effect.

13.6 Optimizing Input Signals with Signal Enhancement and Speckle Tracking

For a scan it is desirable to achieve similar signal quality - i.e. similar noise level - for all scan points. This is, however, made difficult by two problems:

- The speckle nature of the back scattered light
- Differing surface qualities, curved surfaces as well as varying distances for different scan points

The Signal Enhancement and Speckle Tracking functions (Acquisition Settings dialog, SE page) remedy or reduce these problems. This is achieved as follows:

- The noise level is estimated for each individual measurement. In the PSV-3D the worst noise level in the Cartesian coordinate directions x, y, or z is used for the estimate. If averaging is switched on (General page), the individual measurements are averaged and weighted according to their noise level. Measurements with low noise level are weighted higher. The noise level of the averaged spectrum is estimated.
- During a scan the noise level of each scan point is compared with previous scan points. If the noise level is relatively high, additional averages are carried out until the typical noise level is achieved (up to five times the manually set averages). If speckle tracking is activated, the position of the beam is slightly altered (typically 50 µm at 1 m stand-off distance) between the individual measurements. At the end of this procedure a scan point with a relatively low noise level receives the Optimal status. Other scan points receive the Valid status.
- At the end of a scan the signal quality of all scan points is compared. Few scan points may change their status.

With Remeasure activated on the General page, all non-optimal scan points are automatically remeasured (up to ten times) after a scan is finished. The remeasurement starts at scan points with a noise level immediate to optimal scan points.

ⓘ You can stop a measurement any time. Stopped measurements are evaluated and saved in exactly the same way as completed measurements.

13.7 Overlap

Depending on the hardware configuration, in the FFT measurement mode you can set overlap (Acquisition Settings dialog, Frequency page). The software only takes the overlap setting into account for measurements with averaging and without trigger. With overlap, the time for recording the data is reduced.

Principle

The measured time signal is cut into blocks for the FFT. With an overlap of zero, the blocks are right next to each other. If the value is greater than zero, the blocks overlap by this amount. Thus the software records less samples for a measurement.

Example

The following settings have been selected:

- 3 averages
- 100 FFT lines, i.e. 256 samples per block

The software calculates the FFT for this measurement from the blocks as detailed in the following table.

Table 13.1: Calculation of FFT blocks with and without overlap

FFT	FFT	FFT
1st block	1st to 256th sample	1st to 256th sample
2nd block	257th to 512th sample	129th to 384th sample
3rd block	513th to 768th sample	257th to 512th sample

In this example, only 512 samples are acquired with 50% overlap instead of 768 samples without overlap.

If you are measuring with overlap, pay attention to the following:

- The noise reducing effect of the averaging is less, particularly if you use the Rectangle or Tapered Hanning window function.
- For bandwidths above approximately 5 kHz, the time for a measurement is determined by the computing time. In this case, overlap does not give you a time advantage.
- If Signal Enhancement is active, the probability of getting more scan points with the Optimal status in the same time is higher. The best setting in this case is 50% overlap and twice the number of averages than you would set without overlap.

14 Working with 3D Geometries

The PSV Software supports three dimensional geometries for the scan points. This means that in addition to the two dimensional position on the live video image a three dimensional coordinate in space is assigned to every scan point. After the vibration data has been measured at the scan points it can be displayed on the three dimensional geometry of the object under investigation.

3D geometries can be used in the PSV-1D, for the PSV-3D, 3D geometries are mandatory. SECTION 14.1 explains the reason. In SECTION 14.2 you will find the functional principles of 2D and 3D alignment. The different coordinate systems are described in SECTION 14.3. The possibilities of obtaining a 3D geometry in the PSV software can be seen in SECTION 14.4. In SECTION 14.5 you can see how 1D and 3D data is visualized. Finally, SECTION 14.6 gives you some helpful hints on working with 3D geometries.

14.1 Why Using a 3D Geometry?

For PSV-3D systems using 3D geometry is mandatory. The software needs to know the positions of the three scanning heads in the coordinate system of the object under investigation to be able to calculate the directions of the laser beams at the current scan point. Only with this information the vibration components measured in the directions of the laser beams can be transformed to 3D data with the vibration components in x, y and z direction.

When using a 3D geometry the software knows the position of the scanning head in the coordinate system of the object under investigation. For every individual scan point the distance to the scanning head can be calculated. This can be used by the PSV software to automatically focus the laser beam at the current scan point. Before, select Scan > Assign Focus Automatically in acquisition mode.

Knowing the position of the scanning head and the 3D coordinates of the scan points allows for calculating the scanning mirror angles such that the laser beam precisely hits the scan points in space. Especially for objects that are not flat but quite deep this method is superior to the method used to position the laser beam by using the positions on the live video image.

If you want to measure at nodes for example of a finite element model of your object you can import the coordinates of the nodes into the PSV software and measure exactly at these points. Comparing measurement results with calculation results of the finite element software becomes much easier when you use the same positions of the scan points and the finite element nodes.

Using a common 3D coordinate system for a sequence of measurements from different positions and/or directions of the scanning head allows for combining the measurement results into a single result on the combined geometry of the object under investigation.

14.2 Alignment Methods

The laser beam is deflected in the scanning head by mirrors controlled by the PSV software. There are two different procedures that allow for positioning the laser beam on the scan points: standard alignment (2D) and 3D alignment. Both methods allow to calculate the mirror angles for a given scan point either from its position on the live video image (standard alignment) or from its 3D coordinates (3D alignment).

Standard alignment (2D)

Using this function the software can position the laser beam if you click on a position on the live video image. Moreover the laser beam is able to follow the mouse cursor. Scan points can be defined on the live video image.

During standard alignment a sequence of positions of the laser beam are marked on the live video image. After setting an alignment point at the laser spot on the live video image, the beam is moved to a different position and the next alignment point is marked by clicking the position. During this process the software stores a pair of values for every alignment point: the position on the live video image (expressed in video coordinates) and the angles of the horizontal and vertical scanning mirrors. From these pairs of values the software calculates the parameters of a regression function that maps video coordinates to angles and vice versa.

For three dimensional objects this works well only if the camera is positioned very near to the scanning mirrors compared to the distance to the object. Otherwise the different view positions of scanning head and camera lead to parallax effects that can not be compensated by the standard alignment function. For objects on which not all points have the same distance to the scanning head, deviations between the point marked on the live video image and the laser position can occur. Those deviations become larger with greater depth of the object and shorter distance between object and scanning head. They can be avoided using the coaxial unit for the scanning head.

For the PSV-3D the camera used is not parallel to all of the scanning heads. Therefore, the deviations between the point marked on the live video image and the laser position are generally larger. In order to increase the accuracy for the PSV-3D, higher order polynomials are used to calculate the regression if at least 10 points are used for the standard alignment.

3D alignment

Using the 3D alignment, the software can calculate the mirror angles that let the laser beam hit a point in space from the 3D coordinate of this point. This is used by the software during a scan to position the laser beam to points with known 3D coordinates.

During 3D alignment at least 3 alignment points with known distances to the scanning head or at least 4 alignment points for which the distances are not known are used. For every alignment point the 3D coordinates have to be known. The coordinate system of these points defines the coordinate system of the object under investigation (object system). Together with the mirror angles that let the laser beam hit the alignment points, the software calculates the position and direction of the scanning head in the object system.

With the optional geometry scan unit the software is able to measure the distance to the point on the object, where the laser points to. This is used during 3D alignment to measure coordinates of alignment points of a defined type (e.g. origin of the object system, point on a coordinate axis, etc.).

14.3 Coordinate Systems

Video coordinates

The video coordinate system is used to place the scan points on the live video image in acquisition mode and on the video snapshot in presentation mode. The video coordinate system is used internally only.

Scanning head system (PSV-1D)

The scanning head system is a Cartesian coordinate system. The origin of this system is the position where the laser beam with 0°/0° mirror angles would hit the front panel of the scanning head. The orientation of the coordinate system depends on the setting of the direction of the vibrometer channel on the Channels page in the Acquisition Settings dialog. The different orientations of the coordinate axes are summarized in the following table.

Vib Direction	X-axis	Y-axis	Z-axis
+X	to head	up	to the left
+Y	to the right	to head	down
+Z	to the right	up	to head
–X	from head	up	to the right
–Y	to the right	from head	up
–Z	to the left	up	from head

The scanning head system is used if no 3D coordinates are known for the scan points. In this case, you can enter the distance to the object under investigation and the software will calculate the x-, y- and z-coordinates of the scan points from the mirror angles assuming that all scan points are placed on a plane perpendicular to the 0°/0° direction of the laser beam placed at the given distance to the scanning head. As the origin of the scanning head system is placed on the front panel of the scanning head, all scan points will have either the same x-, y- or z-coordinate, depending on the orientation of the coordinate system and therefore on the direction of the vibrometer channel. These 3D coordinates will be displayed for the selected scan point in acquisition and presentation mode below the 2D Point element.

During export of the scan point coordinates to Universal File format, ME'Scope structure files or ASCII files these coordinates will be used as 3D coordinates of the scan points. The coordinate that is equal to the distance to the scanning head will be set to 0 during export for all scan points.

Object system

The object system is the Cartesian coordinate system used for the alignment points during 3D alignment. The coordinates of the alignment points define the object system.

During import of 3D geometries from Universal File format, ASCII format or ME'Scope structure files the 3D coordinates will be interpreted to be in the object system of the current 3D alignment.

During a geometry scan the measured 3D coordinates will be measured in the object system. Therefore you need a valid 3D alignment to perform a geometry scan.

In APS point mode, you can define single scan points. The 3D coordinates are either calculated by measuring the distance to the scan point with the geometry scan unit or via triangulation of the three laser beams that hit this point (only possible with PSV-3D). In both cases, the object system will be used for the 3D coordinates of the defined scan points. Therefore you have to have a valid 3D alignment to define points with 3D coordinates.

The 3D coordinates of the scan points will be displayed in the object system below the element 3D Point when you select a scan point in acquisition or presentation mode. These coordinates will be used during export to Universal files, ME'Scope structure files and ASCII files without further modification.

14.4 Sources of a 3D Geometry

This section describes the different methods for getting a 3D geometry into the PSV software. Every method has its special advantages and drawbacks: Knowing them, you can choose the method suited best for your needs.

Import of geometry data

The import of geometry data is standard for the PSV-3D and optional for the PSV-1D. If you have a valid 3D alignment, i.e. if you have defined an object coordinate system, you can import geometries from Universal File format or ASCII format geometry files or from Vibrant's ME'Scope structure files. To do so, change to APS point mode and in the menu bar select Scan > Import Geometry. You can then navigate to the geometry file and import the geometry from this file.

The geometry definition files contain the 3D coordinates of a set of measurement points and only for Universal File format or ME’Scope structure files lists of connections between these points defining surface and line elements of the geometry. The latter are used to display a surface of the object under investigation in presentation mode.

The 3D coordinates of the measurement points have to correspond to the coordinate system that you have used during 3D alignment. With the 3D alignment the software can position the laser beam to hit the measurement points. The precision of the laser position depends mainly on the quality of the 3D alignment.

The measurement points will be placed on the live video image using the standard alignment. These positions are used for displaying purposes only, they do not influence in any way the measurement results. The precision of the measurement point positions on the live video image (i.e. how well the points are positioned on the corresponding laser spots) depends on the quality of the standard alignment. For deep objects or if the camera position is not at the position of the scanning head (e.g. if you use the center camera of PSV-3D), the quality of the standard alignment might not be satisfactory. The software offers to correct the positions on the live video image automatically. To do so, the software successively positions the laser to every measurement point using the 3D alignment and tries to find the laser spot on the live video image. If it succeeds, this position will be used instead of the position calculated via the standard alignment.

ⓘ During this action no movement should occur in the live video image. Keep the personal from walking through the field of view.

The imported geometry might be larger than the range that can be covered by the live video image and/or the range of the scanning mirrors. Nevertheless, the software will import the complete geometry, placing the points on the live video image if possible. The other points will be visible around the zoomed out live video image. The software marks the points that can not be reached by the scanning mirrors with a color code. These not reachable points will be excluded from the scan automatically. If points are hidden by any parts of the object under investigation, i.e. if there is no direct line of sight from all scanning heads, these points are marked with a different color code. These hidden points will be scanned after a warning message.

Once you have imported a geometry you are free to edit it. You can delete or add points (adding with PSV-1D needs the geometry scan unit) or modify the connections between the points. You can move points around on the live video image and you can modify the 3D coordinates of the selected points to optimize how well the three lasers hit the same point on the object under investigation.

Triangulation

Triangulation is only available for the PSV-3D. If you define your geometry in APS standard or point mode, the PSV software at first has information on the positions of the measurement points on the live video image only. For PSV-3D it is mandatory to have 3D coordinates for the measurement points. Therefore, the software calculates the 3D coordinates from the positions on the live video image by a method called triangulation. Using the standard alignment for all three scanning heads (Top, Left and Right), the scanning mirror angles for every scanning head and for every scan point are calculated. From the scanning head positions and orientations known from the 3D alignment the software calculates the laser beam paths for every scanning head. The 3D coordinate of the point in space where these three beams come closest to each other is used as the 3D coordinate of the respective scan point.

As triangulation uses the standard alignment which is not precise in all cases, the 3D coordinates calculated by this method might not be precise enough. To check this, you can select the scan points one after the other and check if the laser spots hit the object under investigation at the same spot and at the desired position. Here the 3D alignment is used to calculate the mirror angles from the 3D coordinate. These mirror angles are slightly different to the mirror angles calculated from the standard alignment before.

Video Triangulation

The VideoTriangulation function is only available for the PSV-3D with the PSV-S-TRIA option. VideoTriangulation can provide an exact measurement for 3D coordinates. VideoTriangulation needs a 3D geometry to start with, which e. g. can be generated with Triangulation. You will find more detailed information in CHAPTER 10.

Modify 3D coordinates

To modify the 3D coordinates, you can select a single or more points and right click one of the selected points. Select Modify 3D Coordinates in the context menu. The Modify 3D Coordinates dialog appears where you can modify the 3D coordinates of the selected points by changing the coordinates directly or by a slider. Moving the slider will move the 3D coordinate along the direction of the laser beam path. For the PSV-3D the 3D coordinate is moved on a line that corresponds to the average direction of the three laser beam paths. You

can use the slider to minimize the distance between the laser spots on the object under investigation. All 3D coordinates of all selected points will be changed by the same amount when closing the dialog. See also SECTION 10.2 for using VideoTriangulation in the Modify 3D Coordinates dialog.

Geometry scan unit (optional)

With the geometry scan unit, the software can measure distances from the scanning head to any point on the object under investigation that can be reached by the laser. The geometry scan unit contains a laser with a microcontroller that allows for measuring distances. The laser is fed through the optics of the scanning head such that the same scanning mirrors are used to position the geometry laser and the vibrometer laser. The scanning head optics is aligned so that the laser spot of the geometry laser hits the same point on the object under investigation as the vibrometer laser if the same scanning mirror angles are used. You can switch between the two lasers any time in the software by selecting Scan > Geometry Laser.

While the geometry laser is active, you will see a signal strength indicator for the geometry laser in the scanning head control. It is recommended that you check the signal strength for the part of the object under investigation that you want to scan. The indicator should show some signal strength and the OVER LED should be green. If applicable, modify either the filter setting or change the distance of the scanning head to the object. As you might have to change the distance of the scanning head to the object, do this before you perform the standard or 3D alignment because otherwise you would have to repeat the alignments. You can move the laser without a valid standard alignment either with the Position icons in the scanning head control or by pressing and holding down the middle mouse button (mouse wheel) while moving the mouse.

Using the PSV software you can perform a geometry scan. First you have to perform a standard alignment, so that the software can position the laser using the position of the measurement points on the live video image. Then you have to perform a 3D alignment to define the object coordinate system that is used for the 3D coordinates measured during the geometry scan. After this you can define your geometry as usual using one of the APS modes. Having left APS mode, you can start the geometry scan. To do so, select Scan > Geometry Scan. Note that this command is only enabled if you have a valid standard and 3D alignment. The coordinate system of the 3D coordinates is defined by the coordinate system of the alignment points of the 3D alignment (object system).

If you want to check the distance measurement at some points before performing the geometry scan, position the laser to the desired points and select Scan > Geometry Point. The software will measure the distance to the point on the object under investigation and display a dialog were you can read the measured distance and the 3D coordinate of the point in the object system.

During the geometry scan the software positions the geometry laser beam to the scan points calculating the scanning mirror angles from the positions of the scan points on the live video image using the standard alignment. Therefore the precision on how well the positions on the live video image match to the positions on the object depends on the quality of the standard alignment.

ⓘ If you are using a PSV-3D system, we recommend using the video camera of the Top scanning head to display the live video image because this allows for higher quality in the standard alignment of the Top scanning head which is used for the geometry scan.

If the distance measurement fails for some scan points, the software will display a message and offers to repeat the measurement or interpolate the coordinates for the remaining points. During interpolation the distance of a scan point is interpolated from the distances of the neighbors of this point. Then the coordinate is calculated from this distance and the position of the point on the video image. If the point has no neighbors with coordinates, the coordinate remains unmeasured and the dialog is displayed again. If you click Cancel in this dialog, the remaining points will receive the No 3D coordinate point status.

The distance measurement might fail for the following reasons:

- If the amount of light is too much or little, you can try to paint the object, attach a piece of paper to it or find some other means to change the reflective properties of the object's surface. You might even want to remove the specific points from your geometry by converting the geometry to a point geometry and edit it in APS point mode. Repeat the geometry scan after you have done this.
- Not the whole laser spot hits a single part of the object. You might want to convert your geometry to a point geometry by entering the APS point mode and move this point. Then repeat the geometry scan.

If you want to add or remove some points of the geometry you can switch to APS point mode. This will keep the 3D coordinates of the points. You can add new points by positioning the laser on the object and select Scan > Point at Laser Position. The software will automatically perform a distance measurement to this point, calculate the corresponding 3D coordinate and add the point to the geometry.

14.5 3D Geometry and Vibration Data

After you have acquired a 3D geometry you can perform a vibration measurement at the defined scan points. The software will position the vibrometer laser to the measurement points by calculating the scanning mirror angles from 3D coordinates via the 3D alignment. After that you can switch to presentation mode and analyze the vibration data.

There is a 2D and a 3D mode to display the data. In 2D mode the geometry and measurement results are displayed on the live video snapshot taken at the beginning of the vibration measurement scan. You can use color-coding and different view styles to analyze the vibration data.

In 3D mode the vibration data is displayed on a projection of the 3D geometry. Using a virtual camera, the 3D geometry is projected to the view area of the presentation window. By using the mouse and other controls of the software you can move the camera around the object and get an impression of the geometry from different view angles.

In addition to color-coding on the surface of the object, the vibration data is displayed as displacement from the zero position of the geometry. The 3D geometry is deformed by the vibration data. You can display the zero position together with the deformed geometry of the vibration data. To do so, mark the Zero Position check box on the 3D View page in the Display Properties dialog.

During phase animation, the instantaneous values are calculated and the deformed geometry and color-coding are updated accordingly. During time domain animation the deformed geometry and color-coding are updated according to the measurement values corresponding to the current time after trigger.

Please note that the displacements used to visualize the vibration data do not represent real displacements of the object under investigation. You can not only visualize displacement data like this but also data in other physical units like velocity, acceleration or FRFs. Even when you display displacement data the displacement in general is exaggerated by a large number of orders of magnitude. The PSV usually measures displacements in the µm range that are displayed as displacements in the m range.

Visualize one dimensional data (PSV-1D)

The geometry is deformed along the coordinate axis, that corresponds to the direction of the acquisition channel. If e.g. the direction of the channel is –X then positive values are displayed as displacement along the negative x-coordinate axis and negative values along the positive x-coordinate axis. If the vibrometer channel direction is +Z then positive values are displayed along the positive z-coordinate axis and negative values along the negative z-coordinate axis.

ⓘ An angle correction is available to correct the measured vibration data for the incident angle of the laser beam at the surface of the object under investigation, because the vibrometer measures the vibration component along the laser direction. On the Scanning Head page in the Preferences dialog you can activate a correction for the horizontal and vertical scanning directions. The algorithm used for this assumes a planar object surface perpendicular to the 0°/0° direction of the laser beam and a pure out-of-plane vibration. If this is a bad approximation to your object under investigation, you might consider to switch off the angle correction altogether, displaying the uncorrected vibration component along the laser beam.

Visualize three dimensional data (PSV-3D)

For three dimensional data measured with the PSV-3D the color codes correspond to the following value:

$$A = \sqrt{A_x^2 + A_y^2 + A_z^2} \qquad \text{Equation 14.1}$$

A_x, A_y and A_z correspond to the components measured in the corresponding coordinate axis directions of the object coordinate system. You can switch off any combination of components by deselecting the corresponding icons in the toolbar of the presentation window. In this case the corresponding colors are calculated from the remaining active components only. All displayed values are positive, as can bee seen from equation (14.1). The phase display is not available for 3D data. To display single direction components with sign information or phase, switch to 2D display mode and select the corresponding vibrometer channel Vib X, Vib Y or Vib Z.

The position of the deformed geometry is calculated from the individual A_x, A_y and A_z components by adding the scaled components with sign information to the 3D coordinate of the measurement point. Note that only those components are taken into account, that are active. It is not possible to display values that are positive or negative only as a deformed geometry, e.g. it is not possible to display magnitude, magnitude dB or magnitude dB(A) as deformed geometry, because those would be displayed as a misleading displacement towards the positive or negative coordinate axes only.

14.6 Best Practices

This section provides a list of practices that you might find useful when you are working with 3D geometries.

To set up the measurement, proceed as follows:

1. Check the signal strength of the geometry laser.
2. Carry out a standard alignment.
3. Carry out a 3D alignment.
4. Define the geometry.
5. Make a geometry scan.
6. Select Scan > Assign Focus Automatically.
7. Make a vibration measurement scan.

If you have a 3D geometry, using Assign Focus Automatically is the best choice. As the software knows the distances to all scan points it is enough to determine the laser focus values at a very small number of scan points and interpolate the rest of the focus values using the known distances.

If the Assign Focus Automatically command fails, you might want to exclude scan points from being used for the algorithm. To do this, disable these scan points by selecting them, right click one of them and select Disable in the context menu. The Assign Focus Automatically algorithm will then only use the points that are not disabled. Nevertheless, if it succeeds, the determined laser focus values are assigned to all points, including the disabled points. Enable the points you have disabled again, so that they are included in the measurement.

If you are using the geometry scan unit with the PSV-3D, use the video camera in the Top scanning head to perform the standard alignment. This minimizes effects caused by the different view directions and positions of the camera and the scanning head that can not be fully compensated by the standard alignment for deep objects. A better standard alignment quality allows for a better match of the measurement point positions on the live video image to the measurement points on the object.

Stitching

If your object under investigation is too large to be scanned from a single position of the scanning heads or if you want to scan different sides of the object, you can stitch a sequence of measurements. In this case, you have to use a common coordinate system for all measurements, i.e. the reference points of all 3D alignments have to be defined in the very same coordinate system. To achieve this, proceed as follows:

1. Open the Devices page in the Preferences dialog and select the PSV-F-500-3D (1D), PSV-E-401-3D (1D), or PSV-E-400-3D (1D) junction box.
2. Open the Channels page in the Acquisition Settings dialog and deactivate the Vibrometer Left and Vibrometer Right channels.
3. Position the Top scanning head with the geometry scan unit attached at a position where you have a direct line of sight to a large part of your object under investigation.
4. Perform a standard and 3D alignment for the Top scanning head only. For the 3D alignment either use reference points that you have determined before (see below) or define your object coordinate system with one of the coordinate definition modes of the 3D alignment. Use the geometry scan unit during alignment.
5. Mark reference points on the object under investigation. Mark the points with a unique number. For every position of the scanning heads during the later measurements, you have to have at least four reference points that can be directly reached by all scanning heads from their position. It is recommended to use at least five points for optimal quality of the alignment.
6. Switch to APS Point Mode and define scan points at the reference points. Position the laser of the Top scanning head on the reference points and use the Point at Laser Position command at every reference point. Make sure that the indices of the points match those of the marks on the object under investigation. You can change the indices of the points, if necessary. The position of the points on the live video image will not be used, only the 3D coordinates of the points are important.
7. Repeat steps 3 to 6 until you have reference points on all parts of the object under investigation that you want to scan. When you change the 3D alignment in step 4, the software will ask you if the scan point positions on the live video image should be recalculated. Click No, because in this case the positions are irrelevant. You can even move the points out of the way on the live video, if you want to. This will not change the 3D coordinates of the points. When you perform the 3D alignment select the Points with free Coordinates mode and use reference points with known coordinates.
8. Leave APS Point Mode and export the scan points to a Universal file. To do so, in acquisition mode, select File > Export > Universal File.
9. Open the Devices page in the Preferences dialog and select the PSV-F-500-3D, PSV-E-401-3D, or PSV-E-400-3D junction box.
10. Position all scanning heads for the first measurement.
11. Open the 3D Alignment dialog. Deactivate the Auto check box. Import the coordinates of the reference points by clicking Import. A list of the reference points with their unique numbers and their coordinates will be imported into the table at the lower part of the 3D Alignment dialog.

12. Define a new alignment point at one of the reference points marked on the object under investigation. Position all laser beams on this point and assign the laser positions to the alignment point.
13. Select the reference point in the coordinate list and assign its 3D coordinate to the alignment point. Use the unique number of the reference point to identify the entry in the list.
14. Repeat steps 12 and 13 for more reference points.
15. Calculate the 3D alignment, define the geometry and perform the scan.
16. Repeat steps 10 to 15 for all measurements in the measurement sequence.
17. Change to presentation mode and stitch all measurements by creating a new combined file.

If the coordinate definition modes available during 3D alignment with the geometry scan unit do not suite your needs, you might want to do the following: Start with either Origin, Axis, Plane or Three Points on Axes by defining the alignment points. Calculate the 3D alignment. This will enter the actual coordinates of the points into the coordinate list. Switch to Points with free Coordinates mode and modify the coordinates of the alignment points. To do so, click the alignment point in the live video image with the right mouse button, select Points with free Coordinates and modify the coordinates in the appearing dialog. By adding an offset to the coordinates you can e.g. move the origin of the coordinate system by a specific amount.

If you are working with PSV-3D and the geometry scan unit and if you want to use the 3D alignment in Points with free Coordinates mode but the position and/or coordinates are not known precisely, use the points with the free coordinates for the Top scanning head only. Add additional alignment points for the Left and Right scanning heads. You have to position the Top laser to these alignment points, too, because the Top scanning head is used to measure the distances to this points, but during alignment the points are only used for the Left and Right scanning heads. As a result you obtain a 3D alignment where the object system is defined with the same accuracy as the free coordinates you used for the Top scanning head. However, the accuracy of the relative positions of three scanning heads to each other is higher, which is important for the accuracy of the laser positioning and the coordinate transformation during the vibration measurement scan.

If you are using the geometry scan unit to acquire the 3D geometry of the object, first define the raw geometry using standard APS mode. Make sure that all points are well apart from edges of the object under investigation and are fully hit by the geometry laser. Then perform the geometry scan. After this switch to APS point mode and add the more delicate points near to the edges of the object under investigation. Note that you can use the PSV-Z-051 hand set, the PSV-A-PDA, or the PSV Commander (PSV-A-550) to position the laser beam directly at the object and teach in the points. This minimizes the effects of a non perfect standard alignment on the acquisition of the 3D geometry.

If you repeatedly work with the geometry scan unit to measure distances, e.g. while defining alignment points in 3D alignment or teaching in geometry points in APS point mode, switch to the geometry laser by selecting Scan > Geometry Laser. This saves time because now the software does not switch from and to the vibrometer laser any more for every distance measurement.

Use the hand set, the PDA, or the PSV Commander to define alignment points in 3D alignment. With the hand set, the PDA or the PSV Commander you are free to walk around, go to the object under investigation, position the laser beam there and define new alignment points. Using one of these accessories you can position the laser beam more precisely than by looking at the live video image.

Appendix A: Introduction to Complex Numbers

In this appendix you will find the following information on complex numbers:

- Representation of complex numbers - See SECTION A.1 on this.
- Addition and subtraction - See SECTION A.2 on this.
- Multiplication and division - See SECTION A.3 on this.
- The complex conjugate - See SECTION A.4 on this.

A.1 Representation of Complex Numbers

Complex numbers are underlined in this section.

A complex number consists of a real part and an imaginary part.

$$\underline{Z} = \mathrm{Re} + i \cdot \mathrm{Im} \qquad \text{Equation A.1}$$

Re… Real part of $\underline{Z}$

Im… Imaginary part of $\underline{Z}$

i… Imaginary unit, $i = \sqrt{-1}$

As an alternative to the presentation with a real and an imaginary part, complex numbers can also be described with an absolute value and a phase (Euler's formula).

$$\underline{Z} = |\underline{Z}| e^{i\varphi} \qquad \text{Equation A.2}$$

$|\underline{Z}|$… Absolute value of $\underline{Z}$

φ… Phase of $\underline{Z}$

e… Euler's number (e = 2.718…)

The exponential function with an imaginary argument can be replaced by trigonometric functions.

$$e^{i\varphi} = \cos\varphi + i\sin\varphi \qquad \text{Equation A.3}$$

This results in the following:

$$\underline{Z} = |\underline{Z}|\cos\varphi + i|\underline{Z}|\sin\varphi \qquad \text{Equation A.4}$$

$$\mathrm{Re} = |\underline{Z}|\cos\varphi \qquad \text{Equation A.5}$$

$$\mathrm{Im} = |\underline{Z}|\sin\varphi \qquad \text{Equation A.6}$$

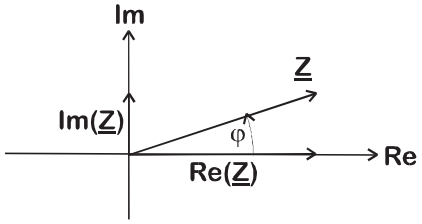


Figure A.1: Graphical presentation of a complex number

The complex plane is a coordinate system with real numbers in x-direction and imaginary numbers in y-direction.

A complex number $\underline{Z}$ is depicted as a pointer in the complex plane. Real part $\mathrm{Re}(\underline{Z})$, imaginary part $\mathrm{Im}(\underline{Z})$, absolute value $|\underline{Z}|$ and phase φ can be taken from the pointer diagram. The phase is 0 if the pointer is in x-direction and 90° if the pointer is in y-direction.

A.2 Addition and Subtraction of Complex Numbers

Addition of complex numbers:

$$\underline{Z}_1 + \underline{Z}_2 = \mathrm{Re}_1 + \mathrm{Re}_2 + \mathrm{i}(\mathrm{Im}_1 + \mathrm{Im}_2) \qquad \text{Equation A.7}$$

Subtraction of complex numbers:

$$\underline{Z}_1 - \underline{Z}_2 = \mathrm{Re}_1 - \mathrm{Re}_2 + \mathrm{i}(\mathrm{Im}_1 - \mathrm{Im}_2) \qquad \text{Equation A.8}$$

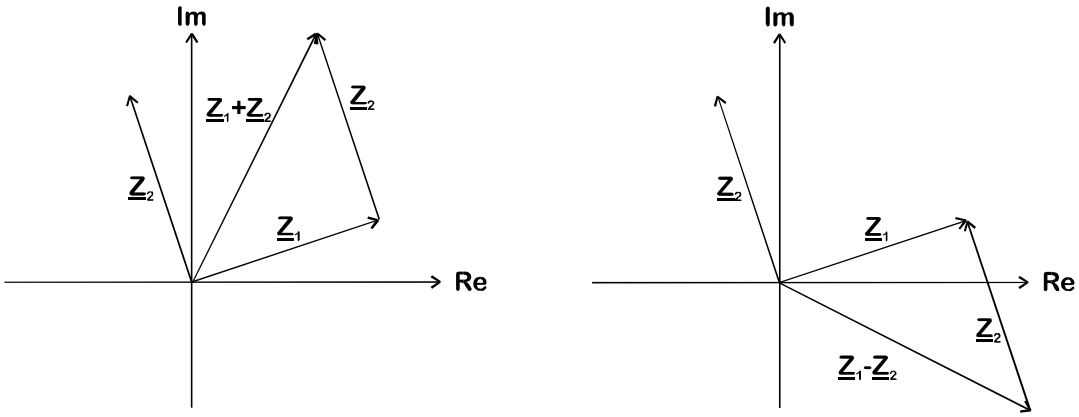


Figure A.2: Addition and subtraction of complex numbers

In the left pointer diagram, two complex numbers are added by vector addition.

In the right pointer diagram, the pointer $\underline{Z}_2$ is subtracted from the pointer $\underline{Z}_1$ by vector subtraction.

A.3 Multiplication and Division of Complex Numbers

Multiplication of complex numbers:

$$\underline{Z}_1 \cdot \underline{Z}_2 = Re_1 \cdot Re_2 - Im_1 \cdot Im_2 + i(Re_1 \cdot Im_2 + Re_2 \cdot Im_1) \qquad \text{Equation A.9}$$

The multiplication is much simpler if the complex numbers are described with an absolute value and phase.

$$\underline{Z}_1 \cdot \underline{Z}_2 = \left|\underline{Z}_1\right| \cdot \left|\underline{Z}_2\right| \cdot e^{i(\varphi_1 + \varphi_2)} \qquad \text{Equation A.10}$$

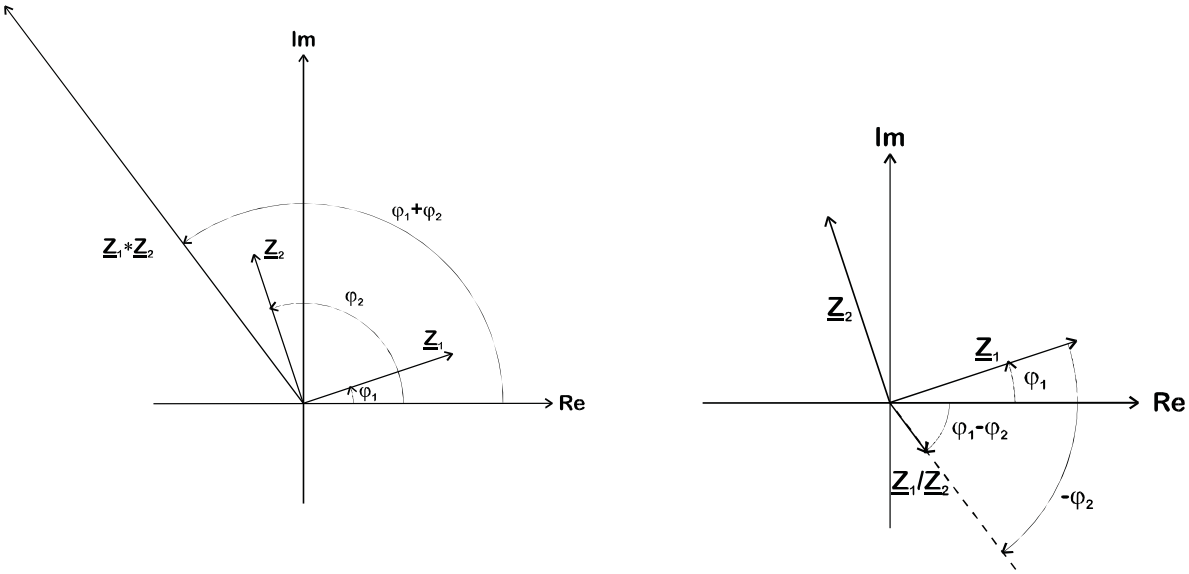


Figure A.3: Multiplication and division of complex numbers

Two complex numbers are multiplied by multiplying the absolute values and adding the phases.

$$\frac{\underline{Z}_1}{\underline{Z}_2} = \frac{\left|\underline{Z}_1\right|}{\left|\underline{Z}_2\right|} \cdot e^{i(\varphi_1 - \varphi_2)} \qquad \text{Equation A.11}$$

Two complex numbers are divided by dividing the absolute values and subtracting the phases.

A.4 The Complex Conjugate

The complex conjugate of $\underline{Z}$ is described as follows:

$$\underline{Z}^* = \mathrm{Re} - i \cdot \mathrm{Im} \qquad \text{Equation A.12}$$

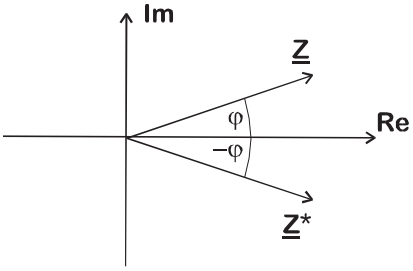


Figure A.4: Conjugated complex of Z

In the pointer diagram the complex conjugate develops through reflection on the real axis.

According to the multiplication rules the following applies:

$$\underline{Z}^* \underline{Z} = |\underline{Z}| \cdot |\underline{Z}| \cdot e^{i(\varphi - \varphi)} = |\underline{Z}|^2 \qquad \text{Equation A.13}$$

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